COMPLETING THE IID BERRY–ESSEEN INEQUALITY AT ESSEEN'S CONSTANT

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We give a computer-assisted completion of the iid Berry–Esseen inequality at Esseen's constant for arbitrary standardized real distributions with finite third absolute moment and every positive sample size. The finite-index argument uses a common coupling of conditional residuals: all incoming binomial count layers are charged against the loss at the current layer. One-sided second-moment comparison bounds the resulting test-function expectations by explicit finite reference laws, while the third-moment cost remains that of the original distribution. The noise, inward support and moment cost are constrained by one shared resource budget. Convexity, lower curvature bounds and finite lower envelopes of valid prices extend finite calculations over continuous parameter domains, including atom activations and closed boundaries. These comparisons complete the five actual-index implications from 26 through 30. Combined in predecessor order with the supporting low-index, bridge and local arguments from the moment-defect framework of Rodziewicz and Kołodziejek, they cover all sample sizes. A Gaussian Peano remainder supplies the initial support radii without index subdivisions. Using the existing heat-smoothing estimate, we start the uniform local induction at 47 from two sharp predecessors. We state the exact global inputs and distinguish their continuous proofs from the rational computations certifying the remaining regions.

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1. The problem and the proof architecture. Let $X_1, X_2, \dots$ be iid real random variables with $\mathbb{E}X_1 = 0$, $\mathbb{E}X_1^2 = 1$, and $\rho = \mathbb{E}|X_1|^3 < \infty$. Put

$$\Delta_n(P) = \sup_{x\in\mathbb{R}} \left| \mathbb{P}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}} \le x \right) - \Phi(x) \right|, \qquad C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The sharp iid conjecture asks whether

$$\sqrt{n}\,\Delta_n(P) \le C_{\mathrm{E}}\rho \quad \text{for every such } P \text{ and every } n \ge 1.$$

Esseen's Bernoulli asymptotic lower bound makes this constant necessary; Schulz proved the complete binomial case [3]. The mathematical issue is the uniform bound for arbitrary real laws, rather than the identification of the candidate constant. An eventual-in-sample-size theorem of He and Cheng [7] gives a uniform upper bound beyond an explicit large threshold. It does not settle the finite range. The moment-defect framework of Rodziewicz and Kołodziejek [8] provides a different route through finite-index localization and propagation. Its public conditional continuation retains finite-index premises. Our comparison argument addresses the remaining five actual-index regions; the proof dependencies needed for the all-index conclusion are made explicit below.

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The finite-index argument below retains a single original law throughout. Its conditional variances, inward support and third absolute moment pay the same budget. The reference laws introduced by moment comparison bound expectations of particular test functions; they do not replace the original law in the statement of the theorem. The proof uses four ranges: the supporting low-index results through 25, the five new implications through 30, the actual-index bridge through 46, and a uniform induction for all $n \geq 47$. The last step uses a support-free Gaussian replacement estimate to discharge the hypotheses of the existing local comparison theorem.

2. Common residuals and moment prices. Fix $0 < p < 1$, write $q = 1 - p$ and $v = pq$, and let B be Bernoulli with parameter p. Conditional on $B = i$, let U have the law of a centered variable U_i, where

$$-O_0 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq O_1, \qquad a_0 + a_1 \leq \Delta < 1. \tag{2.1}$$

Here $O_i > 0$ and $a_i \geq 0$. Put $s_i = \mathbb{E}U_i^2$, $m_i = \mathbb{E}U_i^3$ and $s = qs_0 + ps_1$. We assume that $Y = B + U - p$ is nonpositive on $\{B = 0\}$ and nonnegative on $\{B = 1\}$. Thus $X = Y/\sqrt{v+s}$ has mean zero and variance one. Throughout this section

$$\rho = \mathbb{E}|X|^3, \qquad R_0 = v(1 - 2v)$$

refer to this actual law.

Let $(B_i, \widetilde{U}_i)$ be independent copies of (B, U), and put $S_n = \sum_{i=1}^n (B_i + \widetilde{U}_i)$. For $1 \leq k < n$, write $N = n - k$, $\pi_j = \binom{n}{j} p^j q^{n-j}$, $F_k = \sum_{j=0}^k \pi_j$, and

$$H_r = \sum_{j=k+1}^{k+r} \pi_j, \qquad d = 1 - \Delta.$$

On an auxiliary probability space take independent pairs (U_{0i}, U_{1i}), each with the prescribed marginals, and set $W_j = \sum_{i \leq j} U_{1i} + \sum_{i > j} U_{0i}$. Conditioning on $\sum B_i = j$ gives the same residual law W_j.

LEMMA 2.1 (One coupled staircase). *Let t be real and choose any slope satisfying*

$$\eta \geq \max_{1 \leq r \leq N} \frac{H_r}{rd}.$$

Then

$$\mathbb{P}\{S_n \leq k + t\} \leq F_k + \eta \left\{ t + \mathbb{E}(W_k - t - \pi_k/\eta)_+ \right\}. \tag{2.2}$$

If $h > 0$ satisfies

$$\frac{H_r}{\pi_k} \leq \frac{2rd}{h} + \frac{r^2 d^2}{h^2} \quad (1 \leq r \leq N), \tag{2.3}$$

then

$$\mathbb{P}\{S_n \leq k + t\} \leq F_{k-1} + \frac{\pi_k}{h^2} \mathbb{E}(t + h - W_k)_+^2. \tag{2.4}$$

In particular, writing $V = \mathbb{E}W_k^2 = Ns_0 + ks_1$ gives

$$\mathbb{P}\{S_n \leq k + t\} \leq F_k + \frac{\pi_k}{h^2}(V + t^2 + 2ht). \tag{2.5}$$

When $\pi_{k+1}, \ldots, \pi_n$ decrease, the least legal stop-loss slope is π_{k+1}/d, and it suffices in (2.3) *to check $r = 1$.*

PROOF. The support bounds give $W_{k+r} - W_k \geq -r\Delta$ in this coupling. After discarding losses at counts below k, the CDF minus F_k is at most $\mathbb{E}g(t - W_k)$, where

$$g(y) = -\pi_k \mathbf{1}_{\{y<0\}} + \sum_{r=1}^{N} \pi_{k+r} \mathbf{1}_{\{y \geq rd\}}.$$

For every y, including each jump point, $g(y) \leq \eta y + \eta(-y - \pi_k/\eta)_+$. Taking expectations proves (2.2). Likewise $g(y) + \pi_k \leq \pi_k(y+h)_+^2/h^2$: at the rth incoming jump this is precisely (2.3), and between jumps the right side increases. This proves (2.4); replacing the positive part by the full square gives (2.5). Finally $H_r \leq r\pi_{k+1}$ in the decreasing case. □

For $t < 0$ and $V \leq \overline{V}$, the choice $h = -t + \overline{V}/(-t)$ in (2.5), when legal, gives $F_k - \pi_k t^2/(\overline{V} + t^2)$. All these inequalities hold for closed CDFs and allow arbitrary atoms in the conditional laws.

The moment identity, cubic endpoint inequality and shared budget below come from the framework of Rodziewicz–Kołodziejek [8]; we retain their allocation explicitly for the conditional prices.

LEMMA 2.2 (The actual third-moment budget). *The standardized moment satisfies*

$$\rho(v+s)^{3/2} = R_0 + 3v(s_0 + s_1) - qm_0 + pm_1. \tag{2.6}$$

If $s > 0$, put $\alpha = ps_1/s$. For any positive cap $a_0 + a_1 \leq \delta$,

$$\rho \geq g_\delta(\alpha) := \frac{R_0 + \{3[p + \alpha(q-p)] - \delta\}s + \{(1-\alpha)/\sqrt{q} + \alpha/\sqrt{p}\}^2 s^2/\delta}{(v+s)^{3/2}}. \tag{2.7}$$

The same allocation gives the count variance

$$V_k = s\left\{\frac{N(1-\alpha)}{q} + \frac{k\alpha}{p}\right\}. \tag{2.8}$$

PROOF. Expand $|Y|^3 = (p - U_0)^3$ and $(q + U_1)^3$ in the two groups to obtain (2.6). If Z is centered, $Z \leq a$ and $\mathbb{E}Z^2 = z$, then $\mathbb{E}[(a - Z)(Z + z/a)^2] \geq 0$ yields $\mathbb{E}Z^3 \leq az - z^2/a$. Apply this to U_0 and $-U_1$. Zero caps force the corresponding centered residual and variance to vanish; omit their zero contributions. Cauchy–Schwarz gives

$$-qm_0 + pm_1 \geq -s\{(1-\alpha)a_0 + \alpha a_1\} + \frac{(\sqrt{q}\, s_0 + \sqrt{p}\, s_1)^2}{a_0 + a_1}.$$

Bound the first term below by $-s\delta$ and the denominator above by δ, then use $v(s_0 + s_1) = s[p + \alpha(q-p)]$ to obtain (2.7). Equation (2.8) follows from independence. □

If $p < 1/2$, this floor increases with α for fixed δ. For any positive affine cap $\delta(\alpha)$ it is convex in α, since an affine square divided by a positive affine function is convex. At $s = 0$ both residuals vanish almost surely, and (2.6) gives the Bernoulli moment directly.

The support bounds also imply $s_i \leq O_i a_i$ by averaging $(U_i + \ell_i)(r_i - U_i) \geq 0$ over the corresponding support $[-\ell_i, r_i]$. Suppose an upstream resource bound for this law gives

$$g(a_0 + a_1) + \lambda\{(1-\alpha)\mathfrak{h}_- + \alpha\mathfrak{h}_+\} \leq \mathfrak{R}, \qquad \lambda = ns, \quad g > 0.$$

Its coefficients must be supplied by the applicable resource theorem. Combining it with $a_0 + a_1 \geq s_0/O_0 + s_1/O_1$ yields

$$\lambda\left\{(1-\alpha)\left(\mathfrak{h}_- + \frac{g}{nqO_0}\right) + \alpha\left(\mathfrak{h}_+ + \frac{g}{npO_1}\right)\right\} \leq \mathfrak{R}. \tag{2.9}$$

This necessary plane retains the actual allocation and its common noise.

LEMMA 2.3 (Classical two-moment comparison). *For $a > 0$, $z \geq 0$, let $T_{a,z}$ have mean zero, variance z and atoms $a, -z/a$, with probabilities $z/(a^2+z), a^2/(a^2+z)$; put $T_{a,0} = 0$. If U is centered, $U \leq a$ and $\mathbb{E}U^2 \leq z$, then*

$$\mathbb{E}(U-c)_+^2 \leq \mathbb{E}(T_{a,z}-c)_+^2 \qquad (c \in \mathbb{R}). \tag{2.10}$$

This comparison is preserved by independent addition. The reference expectation increases when either a or z increases. If $U_1, \ldots, U_m$ are independent, centered, bounded above by one common $b > 0$, and have total variance at most V, then

$$\mathbb{E}\left(\sum_i U_i - c\right)_+^2 \leq \mathbb{E}(Z_b(V)-c)_+^2, \qquad Z_b(V) = \sum_{i=1}^m T_{b,V/m,i}, \tag{2.11}$$

where the reference summands are independent.

PROOF. Write $l = -z/a$, allowing $\mathbb{E}U^2 \leq z$ throughout. If $c \leq l$, then $\mathbb{E}(U-c)_+^2 \leq \mathbb{E}U^2 + c^2 \leq z + c^2$, the reference value. If $c \geq a$, both sides vanish. For $l < c < a$, the pointwise majorant

$$(u-c)_+^2 \leq \left(\frac{a-c}{a-l}\right)^2 (u-l)^2 \quad (u \leq a)$$

has a nonnegative quadratic coefficient and equality at both reference atoms. Its expectation increases with $\mathbb{E}U^2$, proving (2.10). This is the squared-positive-part case of Pinelis [2, Lemma 3.2 and Theorem 2.1]. Apply the same comparison to T_{a_1,z_1} with upper bounds $a_2 \geq a_1$ and $z_2 \geq z_1$: this proves both parameter monotonicities without differentiation. Conditioning on the other independent summands proves preservation by addition. Zero variance is the zero law; a centered variable with upper endpoint zero is also zero and may be omitted. Jensen's inequality includes these boundaries without assigning positive variance to $T_{0,z}$. After replacement with common upper endpoint b, Bentkus's variance equalization [1, Lemma 4.5], scaled from $b = 1$, gives (2.11). □

Only test-function expectations are compared in this lemma. The original conditional distributions need not have finite support. For $K(V,c) = \mathbb{E}(Z_b(V)-c)_+^2$ one also has

$$0 \leq K_V \leq 1. \tag{2.12}$$

There is an order proof. Both $K(V,c)$ and $\mathbb{E}(c - Z_b(V))_+^2$ increase with V: for the latter use $-T_{b,z} \overset{d}{=} T_{z/b,z}$ and the two parameter monotonicities. Their sum is $V + c^2$, hence

$$0 \leq K(V_2,c) - K(V_1,c) \leq V_2 - V_1 \qquad (0 \leq V_1 \leq V_2).$$

The finite reference formula is C^1 for $V > 0$, including atom activations, so (2.12) follows. Zero variance follows directly by Jensen and continuity.

For any finite reference sum Z dominating the squared positive parts of W_k, define

$$Q_Z(A) = \inf_{h>0} \frac{\mathbb{E}(Z - A + h)_+^2}{4h}. \tag{2.13}$$

The inequality $x_+ \leq (x+h)_+^2/(4h)$ gives $\mathbb{E}(W_k - A)_+ \leq Q_Z(A)$. Write

$$M_j = \sum_{z_i - A + h > 0} w_i (z_i - A)^j.$$

On a fixed active set the value is $(M_2/h + 2M_1 + M_0 h)/4$. A positive stationary width satisfies $h^2 = M_2/M_0$ and has price $(\sqrt{M_0 M_2} + M_1)/2$; its adjacent activation inequalities

must also hold. The objective is convex in h, with continuous derivative at activations. If $Z \le A$, its infimum is zero. The analogous clipped-square objective and its interior stationary width are

$$M_2/h^2 + 2M_1/h + M_0, \qquad h = -M_2/M_1 \quad (M_1 < 0),$$

subject also to the legal staircase gate.

LEMMA 2.4 (Convex reference prices). *For a fixed reference law Z, the price Q_Z is nonincreasing, convex and 1-Lipschitz in A.*

PROOF. The map $(A,h) \mapsto \mathbb{E}(Z - A + h)_+^2/h$ is a convex perspective, so partial minimization proves convexity of Q_Z. At a positive optimizer, $Q_Z'(A) = -(M_0 + M_1/h)/2 \in [-1,0]$, by $M_1^2 \le M_0 M_2 = M_0^2 h^2$ and $M_0 \le 1$. Convexity extends these bounds to all subgradients. □

3. Phase interpolation and complementary prices. For fixed actual parameters, let

$$R(t) = \mathbb{P}\{S_n \le k + t\} - \Phi\left(\frac{k + t - np}{\sqrt{npq + \lambda}}\right) - c_n\rho, \qquad \lambda = ns, \tag{3.1}$$

where $0 < c_n \le C_{\mathrm{E}}/\sqrt{n}$. In particular ρ and $npq + \lambda$ remain fixed when t changes. The prices of Section 2 majorize this residual; the actual CDF itself need not be differentiable.

LEMMA 3.1 (Semiconvex interpolation). *Let E be continuous on $[a,b]$ and suppose $E(t) + Mt^2/2$ is convex there, with $M \ge 0$. If $E(a) \le L$ and $E(b) \le U$, then*

$$E(a + y) \le L + \frac{U - L}{b - a}y + \frac{M}{2}y(b - a - y), \qquad 0 \le y \le b - a. \tag{3.2}$$

The same conclusion holds if E' is absolutely continuous and $E'' \ge -M$ almost everywhere, or if $E'' \ge -M$ in the sense of distributions. A supremum of functions sharing this semiconvexity constant remains semiconvex wherever finite.

PROOF. Apply the convex chord inequality to $E(t) + Mt^2/2$ and subtract the quadratic. Each of the derivative conditions implies that this function is convex. Supremum preserves convexity. □

For example, keep a fixed comparison law Z and a legal slope $\eta(p)$, and put

$$E(t) = F_k + \eta(p)\{t + Q_Z(t + \pi_k/\eta(p))\} - \Phi\left(\frac{k + t - np}{\sqrt{npq + \lambda}}\right) - c_n\rho.$$

The reference term is convex in phase. The Gaussian contribution has second derivative

$$\frac{(k + t - np)\phi((k + t - np)/\sqrt{A})}{A^{3/2}}, \qquad A = npq + \lambda.$$

Consequently, throughout $p \in [p_l, p_u] \subset (0, 1/2)$, $\lambda \ge \lambda_l$ and $t \in [a,b]$, a valid constant is

$$M = \frac{(np_u - k - a)_+\phi(0)}{\{np_l(1 - p_l) + \lambda_l\}^{3/2}}. \tag{3.3}$$

In particular $M = 0$ for phases $a \ge np_u - k$. The same argument applies to a fixed clipped-square price, whose positive-part-square term is convex in phase. Uniform bounds on the two phase faces therefore control the whole phase interval. Different valid lower bounds on the same actual ρ may be used to estimate those two faces.

Semiconvexity also permits an atom to cross an activation threshold inside a probability interval. If x is C^2 on that interval, then $(x_+)^2$ is C^1, its derivative is absolutely continuous, and almost everywhere

$$((x_+)^2)' = 2x_+x', \qquad ((x_+)^2)'' = 2\mathbf{1}_{\{x>0\}}(x')^2 + 2x_+x''. \tag{3.4}$$

Thus interval bounds with $x_+ \in [0, (\sup x)_+]$ and the indicator in $[0,1]$ give a valid curvature bound across zero. There is no singular derivative at the crossing. In the rest of this section $Z_b(V)$ has n independent summands, so the number m in (2.11) is n. For instance, if only the lowest atom of $Z_b(V)$ can lie below y, the exact identity

$$\mathbb{E}(Z_b(V) - y)_+^2 = V + y^2 - \left(1 + \frac{V}{nb^2}\right)^{-n} (V/b + y)_+^2 \tag{3.5}$$

may be differentiated in this way. The condition that every other atom is at least y is still required. For $n > 1$ this identity also gives a uniform lower variance derivative when $y \le y_{\max} < b$. Put

$$z_* = \frac{2n - \sqrt{2n(n-1)}}{n+1}, \qquad m_* = 1 - \left(\frac{n - z_*}{n - y_{\max}/b}\right)^{n-1} (2z_* - z_*^2).$$

Then $K_V(V,y) \ge m_*$ throughout any region where at most the lowest atom is clipped. If none is clipped, $K_V = 1$. Otherwise, with $r = 1 + V/(nb^2)$ and $z = (V/b + y)/(br) \in [0,1]$, differentiation gives

$$K_V = 1 - \left(\frac{n - z}{n - y/b}\right)^{n-1} (2z - z^2).$$

The denominator is positive, and the subtracted expression increases with y. Its derivative in z has the sign of $2n - 4nz + (n+1)z^2$, whose unique zero in $[0,1]$ is z_*. This proves the bound, including activation boundaries by continuity.

LEMMA 3.2 (A finite lower envelope of phase prices). *Suppose $R(t) \le E_i(t)$ for $1 \le i \le r$, and that on $[0, H_i]$ the function E_i is semiconvex with constant M_i and has endpoint upper bounds L_i, U_i. Choose $M \ge \max_i M_i$ and $0 < H \le \min_i H_i$, and write*

$$A_i = L_i, \qquad B_i = \frac{U_i - L_i}{H_i} + \frac{MH_i}{2}, \qquad q_i(t) = A_i + B_i t - \frac{M}{2}t^2.$$

Then, throughout $[0, H]$,

$$R(t) \le q(t) := \min_i (A_i + B_i t) - \frac{M}{2}t^2. \tag{3.6}$$

The maximum of q is attained among $0, H$, the vertices B_i/M when $M > 0$, and the intersections $(A_j - A_i)/(B_i - B_j)$ when $B_i \ne B_j$, retaining only points in $[0, H]$.

PROOF. Lemma 3.1 gives $E_i \le q_i$ on its own interval. Their minimum therefore majorizes the same actual residual. Between consecutive intersections of the affine parts, q agrees with one concave quadratic. Its maximum is at a vertex or a segment endpoint. □

No mixing parameter is needed. The old fixed-mixture certificates remain valid because, for every fixed probability vector (θ_i),

$$\min_i q_i(t) \le \sum_i \theta_i q_i(t).$$

Thus any certified upper bound for such a mixture also bounds the lower envelope. With rational face bounds and curvature, every point in the finite candidate set and every value evaluated there is rational. An estimate using a coefficient $c_i \le c_n$ already majorizes the residual with coefficient c_n, since $\rho \ge 0$. Thus the directed lower coefficients in the finite calculation need no alignment cost when the target is $c_n = C_{\mathrm{E}}/\sqrt{n}$.

LEMMA 3.3 (A short CDF bridge). *For the same actual law in* (3.1)*, suppose* $npq + \lambda \ge A_* > 0$ *and* $R(b) \le U$*. Then, for every* $t \le b$*,*

$$R(t) \le U + \frac{\phi(0)}{\sqrt{A_*}}(b - t). \tag{3.7}$$

PROOF. The actual closed CDF is nondecreasing. Hence $R(t) \le R(b) + \Phi((k + b - np)/\sqrt{npq + \lambda}) - \Phi((k + t - np)/\sqrt{npq + \lambda})$. Bound the last difference by the maximum normal density. □

Only the price at b is needed in this bridge. Its comparison law, activation conditions or curvature estimates need not extend to the smaller phases. The fixed actual moment and variance, and monotonicity of the actual CDF, suffice.

4. Continuous envelopes for the finite certificates. The finite calculations use two envelopes. The first freezes a moment-comparison law on an entire parameter box. The second keeps the same noise in the comparison variance and moment cost until a derivative test licenses its elimination. The tests below are obligations of each certificate, rather than consequences asserted for every parameter choice.

PROPOSITION 4.1 (A fixed-reference box). *Let* $p \in [p_l, p_u] \subset (0, 1/2)$ *and* $\ell \le \lambda \le u$*, with* $\nu = np_l(1 - p_l) + \ell > 0$*. Suppose a reference expression* $A(p, t)$*, independent of the actual noise, and a differentiable moment floor* $g(p, \lambda)$ *give*

$$R(t) \le A(p, t) - \Phi\left(\frac{k + t - np}{\sqrt{npq + \lambda}}\right) - c_n g(p, \lambda).$$

If $g_\lambda \ge b$ *throughout the box, then*

$$R(t) \le A(p, t) - \Phi\left(\frac{k + t - np}{\sqrt{npq + \ell}}\right) - c_n g(p, \ell) + J(t), \tag{4.1}$$

$$J(t) = (u - \ell)\left\{\frac{(k + t - np_l)_+ \phi(0)}{2\nu^{3/2}} - c_n b\right\}_+ .$$

If $A(p, \cdot)$ *is convex, the upper function has the phase semiconvexity constant in* (3.3) *with* $\lambda_l = \ell$*.*

PROOF. For $\mu = k + t - np$, differentiation of the explicit surrogate gives

$$\frac{\partial}{\partial\lambda}\left\{-\Phi\left(\frac{\mu}{\sqrt{npq + \lambda}}\right) - c_n g(p, \lambda)\right\} = \frac{\mu\phi(\mu/\sqrt{npq + \lambda})}{2(npq + \lambda)^{3/2}} - c_n g_\lambda.$$

Its upper bound is the expression in braces defining J. Integrate from ℓ to the actual λ. The nonnegative correction J is convex in t, so only the Gaussian term contributes to the lower phase-curvature bound. □

Here A may be the current-count price

$$A(p,t) = F_k + \eta(p)\{t + Q_Z(t + \pi_k/\eta(p))\},$$

or a fixed reflected-square reference price. For example, upper bounds a, z_0, z_1 on the first group's inward support and the two conditional variances give

$$Z = \sum_{i=1}^{n-k} T_{a,z_0,i} + \sum_{i=1}^{k} T_{O_1,z_1,i}.$$

Lemma 2.3 makes this one law valid throughout the box. The necessary resource plane (2.9) is retained when bounding $\lambda(1-\alpha)$ and $\lambda\alpha$ for z_0, z_1. An inward cap and an allocation lower bound similarly give

$$a_0 \le \delta - \frac{\alpha\lambda}{npO_1}.$$

These relations keep the variance and support corners within a single proved resource budget.

The probability direction can also be bounded without a probability mesh. At a fixed phase, put $a(p) = t + \pi_k(p)/\eta(p)$. For the term $\eta Q_Z(a)$, a lower second-derivative bound is supplied by

$$\eta'' Q_Z(a) + 2\eta' Q_Z'(a)a' + \eta Q_Z'(a)a''. \tag{4.2}$$

The omitted term is $\eta Q_Z''(a)(a')^2 \ge 0$, interpreted weakly at an activation point. The remaining binomial, Gaussian and moment-floor derivatives are explicit. Bounds on these expressions and on $-1 \le Q_Z' \le 0$ give a uniform probability chord by Lemma 3.1. Face bounds obtained from different moment estimates are combined through the same fixed-reference envelope with the actual law's original moment and variance.

PROPOSITION 4.2 (A coherent reflected-square envelope). *Let $j = k+1 < n$ and fix a legal square width h for reference count j. Suppose the centered reflected residual summands have one common upper bound $b > 0$. Set*

$$V_j = \frac{\lambda}{n}\left\{\frac{(n-j)(1-\alpha)}{q} + \frac{j\alpha}{p}\right\}, \qquad K(V,y) = \mathbb{E}(Z_b(V) - y)_+^2.$$

Let $G(p,\lambda,\alpha)$ be a valid actual-moment floor, convex in $\alpha \in [0,1]$ and differentiable there. A positive affine inward cap in (2.7) *supplies such a floor. Define*

$$E(p,\lambda,\alpha,t) = F_k + \frac{\pi_j}{h^2}K(V_j, 1-t-h) - \Phi\left(\frac{k+t-np}{\sqrt{npq+\lambda}}\right) - c_n G(p,\lambda,\alpha). \tag{4.3}$$

Suppose, on the whole stated probability, noise and phase domain,

$$c_n G_\alpha(p,\lambda,0) \ge \frac{\pi_j \lambda (j-np)_+}{npq\,h^2}, \tag{4.4}$$

and

$$\frac{\pi_j(n-j)}{nqh^2}K_V(V_j(p,\lambda,0), 1-t-h) + \frac{(k+t-np)\phi((k+t-np)/\sqrt{npq+\lambda})}{2(npq+\lambda)^{3/2}} - c_n G_\lambda(p,\lambda,0) \ge 0. \tag{4.5}$$

Then for every actual $\lambda \in [\ell, u]$ and allocation,

$$R(t) \le E(p,\lambda,\alpha,t) \le E(p,u,0,t). \tag{4.6}$$

The zero-noise boundary is included by continuity.

PROOF. Apply (2.4) at count j and phase $t-1$, then apply (2.11) to the reflected residual sum. This proves the first upper bound with the same actual third moment, replaced only by its valid lower floor G.

By (2.12), $0 \le K_V \le 1$, and $\partial_\alpha V_j = \lambda(j-np)/(npq)$. Convexity gives $G_\alpha(p,\lambda,\alpha) \ge G_\alpha(p,\lambda,0)$. Thus (4.4) implies $E_\alpha \le 0$. On the formal zero-allocation face, (4.5) is exactly $\partial_\lambda E(p,\lambda,0,t) \ge 0$. Integrating the two monotonicities proves (4.6). □

Zero allocation here is an endpoint of the majorizing function; it need not describe an admissible original law. The first elimination uses the universal bound $0 \le K_V \le 1$. Only afterwards is a one-clip formula needed to evaluate the formal face or obtain the sharper lower derivative in (4.5). Its next-atom condition is checked on that face. The resulting explicit function $E(p,u,0,t)$ is bounded in probability and phase by the weak-curvature rules of Section 3.

COROLLARY 4.3 (An allocation interval). *In the same reflected-square setting, retain one allocation interval $[\alpha_-,\alpha_+]$. Suppose the moment floor G is nondecreasing in allocation throughout the domain. For either endpoint $\xi \in \{\alpha_-,\alpha_+\}$ define*

$$E_\xi(p,\lambda,t) := F_k + \frac{\pi_j}{h^2}K(V_j(p,\lambda,\xi),1-t-h) - \Phi\left(\frac{k+t-np}{\sqrt{npq+\lambda}}\right) - c_n G(p,\lambda,\alpha_-). \tag{4.7}$$

Then, throughout the original probability interval, including a crossing of $j=np$,

$$R(t) \le \max\{E_{\alpha_-}(p,\lambda,t), E_{\alpha_+}(p,\lambda,t)\}. \tag{4.8}$$

If $j \ge np$ everywhere, only $\xi=\alpha_+$ is needed; if $j \le np$ everywhere, only $\xi=\alpha_-$ is needed. If $\partial_\lambda E_\xi \ge 0$ on $[\ell,u]$ for every retained branch, replace λ by u in the resulting maximum.

PROOF. The variance V_j is affine in allocation, so its maximum on an interval occurs at an endpoint. Since K increases with variance and the moment cost is at least $G(p,\lambda,\alpha_-)$, (4.8) follows. The derivative $\partial_\alpha V_j = \lambda(j-np)/(npq)$ selects a single endpoint when its sign is fixed. Each retained explicit branch keeps the same actual λ and can be increased to u under its own noise derivative condition. Zero noise is included by continuity. □

The noise gate in this corollary is explicit. With $\nu_\xi = \{(n-j)(1-\xi)/q + j\xi/p\}/n$, it is

$$\frac{\pi_j\nu_\xi}{h^2}K_V(\nu_\xi\lambda,1-t-h) + \frac{(k+t-np)\phi((k+t-np)/\sqrt{npq+\lambda})}{2(npq+\lambda)^{3/2}} - c_n G_\lambda(p,\lambda,\alpha_-) \ge 0.$$

Probability and phase bounds are also applied to each explicit branch before taking their maximum. A single branch across a crossing is only an auxiliary expression; the maximum in (4.8) gives the actual residual bound. Both allocation rules use one endpoint test. The affine-cap moment floor is convex by Section 2, so $G_\alpha(p,\lambda,\alpha_-) \ge 0$ implies $G(\alpha) \ge G(\alpha_-)$. The stronger endpoint payment in (4.4), with 0 replaced by α_-, instead makes the full price nonincreasing. To evaluate that endpoint, write $\delta = A - \lambda(H + r\alpha) > 0$, $B = (1-\alpha)/\sqrt{q} + \alpha/\sqrt{p}$ and $B_\alpha = 1/\sqrt{p} - 1/\sqrt{q}$. Direct differentiation gives

$$G_\alpha = \frac{s}{(v+s)^{3/2}}\left\{3(q-p) + \lambda r + s\left(\frac{2BB_\alpha}{\delta} + \frac{\lambda r B^2}{\delta^2}\right)\right\}. \tag{4.9}$$

For $p \in [p_l, p_u] \subset (0, 1/2)$, $0 \le \lambda \le L$, $\delta \ge \delta_* > 0$ and $B \le B_*$, a lower bound on the braced quantity is

$$3(1 - 2p_u) + L \min(0, r) + \frac{L^2 \min(0, r) B_*^2}{n\delta_*^2}. \tag{4.10}$$

This is an optional sufficient bound for the endpoint test. The omitted $2BB_\alpha$ term is nonnegative, so a positive bound licenses the choice for either sign of r. Zero noise again follows by continuity.

5. Global inputs and their dependencies. For a standardized real law P with finite $\rho(P) = \mathbb{E}|X|^3$, write

$$R_n(P) = \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}, \qquad C_n = \sup_P R_n(P).$$

The supremum is over all such laws, without a support restriction. We use the classical iid moment envelope [6]

$$R_n(P) \le T(\rho) := \min\{.469,\ .3322(1 + .429/\rho),\ .3031(1 + .646/\rho)\}. \tag{5.1}$$

All terminating decimals denote rational numbers.

We include the variational starting point and contact argument; compare He–Cheng [7, Lemmas 6.1 and 6.3, equations (12)–(15)].

LEMMA 5.1 (A contrary extremizer). *If $C_n \ge C_{\mathrm{E}}$, the supremum defining C_n is attained by a standardized law P with $\rho(P) < 2$. After reflection if necessary, there is a finite H such that*

$$F_n(H) - \Phi(H/\sqrt{n}) = C_n \rho(P)/\sqrt{n}, \qquad F_n(H) = P^{*n}((-\infty, H]).$$

PROOF. The classical moment envelope in (5.1) gives $C_n \le .469 < 1/2$ and $T(2) \le .3031(1 + .646/2) = .4010013 < C_{\mathrm{E}}$. Thus a maximizing sequence whose scores tend to $C_n \ge C_{\mathrm{E}}$ eventually has third moments below 2. Tightness and this third-moment bound give a weakly convergent subsequence $P_j \Rightarrow P$ whose first two moments remain $0, 1$. Pass also to $\rho(P_j) \to \rho_* \in [1, 2]$, so $\rho(P) \le \rho_*$.

Reflection converts a negative discrepancy into a limit of positive closed discrepancies. Choose corresponding thresholds H_j with an error tending to zero. These thresholds stay bounded: Chebyshev's inequality and the normal tail bound make the discrepancy tend uniformly to zero as $|H_j| \to \infty$. Passing to $H_j \to H$, the closed-set inequality for weak convergence gives

$$F_n(H) - \Phi(H/\sqrt{n}) \ge C_n \rho_*/\sqrt{n}.$$

Maximality and $\rho(P) \le \rho_*$ force equality and $\rho(P) = \rho_*$. The moment envelope then gives $\rho(P) < 2$.

For the contact condition, put $z = H/\sqrt{n}$ and $m = \mathbb{E}[X|X|]$. Perturb the common marginal to $Q_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_x$. Write $\mu_\varepsilon, \sigma_\varepsilon^2$ and β_ε for its mean, variance and centered third absolute moment. The threshold H stays fixed in these original coordinates. By the definition of C_n,

$$J(\varepsilon) = Q_\varepsilon^{*n}((-\infty, H]) - \Phi\left(\frac{H - n\mu_\varepsilon}{\sigma_\varepsilon\sqrt{n}}\right) - \frac{C_n\beta_\varepsilon}{\sqrt{n}\,\sigma_\varepsilon^3} \le 0 = J(0).$$

The right derivative at zero therefore gives

$$\mathcal{D}(x) = n\{F_{n-1}(H-x) - F_n(H)\} + \phi(z)\left\{\sqrt{n}x + \frac{z}{2}(x^2-1)\right\}$$
$$- \frac{C_n}{\sqrt{n}}\left\{|x|^3 - \frac{3}{2}\rho x^2 - 3mx + \frac{1}{2}\rho\right\} \le 0. \tag{5.2}$$

Here F_0 is the CDF of the point mass at zero when $n = 1$. Indeed the perturbed mean and variance have derivatives x and $x^2 - 1$, and the centered third absolute moment has derivative $|x|^3 - \rho - 3mx$. The convolution derivative is exactly $n\{F_{n-1}(H-x) - F_n(H)\}$, preserving the common marginal. The integral of $\mathcal{D}$ under P is zero. Since $\mathcal{D}$ is upper semicontinuous, it is zero at every point of the support, including atoms.

□

The heat-smoothing estimate below is already present in Rodziewicz–Kołodziejek [8, Section 15].

LEMMA 5.2 (A coarse Gaussian entry). *For $n \ge 2$,*

$$C_n \le C_{n-1}\sqrt{\frac{n}{n-1}} + \frac{\sqrt{n}}{15(n-1)^{3/2}} + \frac{2\sqrt{n}}{25(n-1)^2}.$$

In particular, $C_{n-1} \le C_{\mathrm{E}}$ and $n \ge 26$ imply

$$n(C_n - C_{\mathrm{E}}) < \frac{3}{10}, \qquad C_n < .422. \tag{5.3}$$

PROOF. Put $v = n - 1$ and $F_v(t) = \Phi(t/\sqrt{v})$. Conditioning on the last summand, replacing the first v summands by a Gaussian costs at most $C_v\rho/\sqrt{v}$. Taylor expansion in the last standardized summand gives

$$\left|\mathbb{E}F_v(H - X) - F_v(H) - \tfrac{1}{2}F_v''(H)\right| \le \frac{\phi(0)\rho}{6v^{3/2}} < \frac{\rho}{15v^{3/2}}.$$

The heat identity $\partial_v F_v = F_v''/2$ gives the matching expansion in the variance variable, with error at most $\|\phi'''\|_\infty/(8v^2) < 2/(25v^2)$. Here Gaussian Fourier inversion gives $\|\phi'''\|_\infty \le 2/\pi < 16/25$. Combine the errors, multiply by $\sqrt{n}/\rho$, use $\rho \ge 1$, and take the threshold and law suprema. This is the elementary heat-smoothing estimate also used in the supporting global route. For the last assertion put $x = n/(n-1)$. Under the sharp predecessor,

$$n(C_n - C_{\mathrm{E}}) \le \frac{C_{\mathrm{E}}x}{\sqrt{x}+1} + \left(\frac{1}{15} + \frac{2}{25\sqrt{n-1}}\right)x^{3/2}.$$

When $n \ge 26$, $x \le 26/25$, $\sqrt{x} < 51/50$ and $C_{\mathrm{E}} < .41$. Also $x/(\sqrt{x}+1) \le (3x+1)/8$, since $(\sqrt{x}-1)^2(3\sqrt{x}+1) \ge 0$. Consequently the right side is less than

$$\frac{41}{100}\frac{3(26/25)+1}{8} + \left(\frac{1}{15} + \frac{2}{125}\right)\frac{26}{25}\frac{51}{50} = \frac{747107}{2500000} < \frac{3}{10}.$$

Also $C_{\mathrm{E}} + 3/(10 \cdot 26) < .422$. In particular, the sharp bound $C_{29} \le C_{\mathrm{E}}$ already implies $C_{30} < C_{\mathrm{E}} + .01$, the coarse premise of input (II) below. The sharp index-30 conclusion still needs its own proof. □

The finite-index completion uses the following three conclusions of the supporting proofs.

PROPOSITION 5.3 (Global inputs). *The low-index companion, actual-index bridge and uniform local comparison give the following statements. Put $\delta_n = n(C_{n-1} - C_n)_+$.*

(I) $C_j \le C_{\mathrm{E}}$ *for* $1 \le j \le 25$*;*
(II) *if* $C_{30} < C_{\mathrm{E}} + 1/100$*, then* $C_j \le C_{\mathrm{E}}$ *for every* $31 \le j \le 46$*;*
(III) *for every integer* $n \ge 47$*, the conditions*

$$0 \le n(C_n - C_{\mathrm{E}}) \le 7/10, \qquad \delta_n = 0, \qquad C_n < .4247, \qquad C_{n-1}, C_{n-2} < .4253$$

cannot hold simultaneously.

This is the zero-allowance case of the original local exclusion; the least-failure argument below supplies exactly this condition. The exclusion (III) is uniform in n*; its premises will be discharged by the Gaussian entry and predecessor induction.*

The precise companion statements and their continuous proof sources are identified in Appendix E.

The classical inputs are Prawitz's smoothing inequality [4] in the form given by Shevtsova [5, Lemma 4.1, p. 263], Shevtsova's finite iid moment bounds [6], and Schulz's strict bound for every binomial law [3, Theorem 1]. In particular, the moment envelope (5.1) gives $T(\rho) \le 819/2000 < C_{\mathrm{E}}$ for $\rho \ge 7361/4000$, excluding the entire larger-moment ray. Prawitz smoothing is applied at each actual integer, retaining its endpoint terms and infinite Gaussian tail. The contact, moment-defect, resource and convolution-count reductions are proved in the companions; they are not additional unnamed literature assumptions.

Lemma 5.1 supplies the common starting point on a contrary branch, before any use of higher moments. At this same law,

$$n(R_{n-1}(P) - C_n)_+ \le n(C_{n-1} - C_n)_+. \tag{5.4}$$

Thus a global predecessor bound licenses contact allowances throughout the argument. A score bound restricted to a moment band is never substituted for that predecessor.

For (I), the first four direct bounds are

$$C_1 \le \tfrac{2}{5}, \quad C_2 \le \tfrac{2\sqrt{2}}{7}, \quad C_3 \le \tfrac{59\sqrt{3}}{250}, \quad C_4 \le .4097 < C_{\mathrm{E}}.$$

The companion compositions use the unrestricted predecessor in order through 24. At 25, a support-free Fourier bound excludes $\rho \ge 6/5$; the two lower bands use the global $C_{24} < .419$ and their complete resource/count covers. Their continuous reductions, unseeded resource steps and closed arithmetic covers are the precise inputs listed in Appendix E. The fifth-index proof is also given separately in [9]; subsequent source developments are collected in [10].

For (II), the coarse bound at 30 gives allowance $31/100$ at a contrary index-31 extremizer. The eight closed moment bands have actual-31 support, resource and count proofs. The bridges through 34 keep $mD_m \le 31d$, acquire support at each actual m, and use that integer's binomial law. Subsequent implications proceed through 35–39 and 40–46, retaining both deleted predecessors where required. Thus the bridge starts from the stated coarse premise without any monotonicity assumption on C_n.

For (III), the moment envelope gives $\rho < 7361/4000$ at a contrary extremizer. Corollary B.3 then gives radius 3.84 directly from contact and the deleted-predecessor allowance. It also supplies the two left radial contractions and the central radius-2.5 contraction, in their original acquisition order. Thus the six initial index-band and subsequent radial support calculations are replaced by the same cubic bound. The bounded-threshold coordinate-strip exclusions retain their original proofs. The resource and confinement schedule then supplies 45 central and 38 left probability intervals, covering $[.304, .47]$ after all complementary intervals have been excluded. Their resources retain the actual weighted quantities nD and $n\mathbb{E}U^2$. The complete local comparison is uniform for every $n \ge 47$. At positive noise it gives a strict loss

relative to this law's own Bernoulli score; at zero noise Schulz's strict binomial theorem applies directly. The acquisition includes both deleted-score caps and the full excess interval $0 \le n(C_n - C_{\mathrm{E}}) \le .7$.

PROPOSITION 5.4 (Induction from two sharp predecessors). *Input (III) and* $C_{45}, C_{46} \le C_{\mathrm{E}}$ *imply* $C_n < C_{\mathrm{E}}$ *for every integer* $n \ge 47$.

PROOF. If a failure exists, choose the least $n \ge 47$ with $C_n \ge C_{\mathrm{E}}$. Both predecessors satisfy $C_{n-1}, C_{n-2} \le C_{\mathrm{E}}$, so $\delta_n = 0$ and both deleted-score caps in (III) hold. The Gaussian entry gives

$$0 \le n(C_n - C_{\mathrm{E}}) < 3/10 < 7/10, \qquad C_n < .422 < .4247.$$

All premises of (III) are now satisfied, a contradiction. □

The moment-defect method and the original finite subdivisions retain their attribution to Rodziewicz–Kołodziejek [8]. The conditional continuation in Section 11 of that working manuscript retains analytic and historical certificate premises, including the sharp assertions at 25–30. It does not by itself establish (I) or fill the intervening finite range. Inputs (I)–(III) here refer to the companion proofs with their continuous reductions, complete exact calculations and closed-domain compositions. In particular, the proof order is (I), the new actual-index steps 26–30, (II) through 46, and Proposition 5.4. The five finite ceilings in the independent $n \ge 49$ route are unnecessary for this composition.

6. From a global extremizer to a conditional comparison. Fix an actual integer $26 \le n \le 30$ and assume the unrestricted predecessor bound $C_{n-1} \le C_{\mathrm{E}}$. Suppose $C = C_n \ge C_{\mathrm{E}}$, and take the positive closed maximizing pair (P, H) of Lemma 5.1, with $1 \le \rho \le 6/5$. All quantities below belong to this one pair. Write $\beta = \mathbb{E}[X|X|]$, $\gamma = \mathbb{E}X^3$, $M_j = \mathbb{E}|X|^j$, $z = H/\sqrt{n}$ and $\varphi = \phi(z)$. The moment-defect construction and its original finite subdivisions are due to Rodziewicz–Kołodziejek [8]. We spell out the continuous implications needed to interpret the finite rows; the arithmetic verification of a row is a separate obligation.

First, the predecessor bound has a pointwise support consequence. Set $s_n = \sqrt{n/(n-1)}$, $e_n = n(s_n - 1) - 1/2$, and

$$\mathcal{H}_n(H, x) = n^{3/2}\{\Phi((H - x)/\sqrt{n-1}) - \Phi(H/\sqrt{n})\} + nx\varphi + \tfrac{H}{2}(x^2 - 1)\varphi.$$

Multiplying (5.2) by $\sqrt{n}$ and using $F_{n-1}(H - x) \le \Phi((H - x)/\sqrt{n-1}) + C_{n-1}\rho/\sqrt{n-1}$ gives

$$\sqrt{n}\,\mathcal{D}(x) \le \mathcal{H}_n(H, x) + C\{-|x|^3 + \tfrac{3}{2}\rho x^2 + 3\beta x + \rho e_n\}. \tag{6.1}$$

The otherwise present allowance $n\rho s_n(C_{n-1} - C)_+$ is zero. Since $\mathcal{D} = 0$ on the full support, a strictly negative upper bound for the right side on $x \le -R_-$ or $x \ge R_+$ excludes that entire ray, including atoms. The bound must hold for every real H and every moment and score value in the incoming intervals. Thus each support refinement uses the actual integer n and previously established moment intervals. In particular, a support calculation at 25 cannot simply be relabelled at n. For the initial radius no such calculation is needed: Proposition B.2 gives $12/5$ when $\rho \le 11/10$ and $13/5$ when $\rho \le 6/5$, by one Peano-kernel argument for all $n \ge 25$. This precedes all higher-moment and defect refinements.

Here is the physical resource behind the refinement. Put $\sigma=\sqrt{10}$ and $\tau(t)=\mathbb{E}[X\mathbf{1}_{\{X\ge t\}}]$. For every bounded component $g=(a,b)$ of the complement of the closed support, τ is constant there; call the value τ_g. Define

$$J=\sum_g \tau_g(b-a)^2, \qquad J_1=\sum_g \tau_g\int_a^b (t-a)t\,dt,$$

$$J_2=\sum_g \tau_g\int_a^b (t-a)t|t|\,dt, \qquad D=(3+\sigma)\rho-\gamma-3J.$$

All these expressions are finite by the bounded support already proved. For $\nu=\mathcal{L}(H-X_1-\cdots-X_{n-1})$, also put

$$\mathcal{L}=\sum_g \tau_g\int_{[a,b)}(t-a)\,\nu(dt)\ge 0.$$

The half-open interval specifies the convention at atoms.

LEMMA 6.1 (The physical defect). *One has $D\ge 0$. If $D<(\sigma-1)\rho$, there is an actual central gap $(-A_-,A_+)$ with $A_-,A_+>0$. Put*

$$p=\mathbb{P}(X>0),\quad q=1-p,\quad v=pq,\quad \mu=\mathbb{E}X_+=\mathbb{E}X_-,\quad \bar A_-=\mu/q,\quad \bar A_+=\mu/p,$$

and $\Delta_\pm=\bar A_\pm-A_\pm\ge 0$. Let $\theta_\pm=\mathbb{E}[(X-\mathbb{E}[X\mid \operatorname{sgn}X])^2;\ \pm X>0]$ and $\theta=\theta_-+\theta_+$. Then, with $p_*=(4-\sigma)/2$,

$$D\ge \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{v}}(p-p_*)^2$$

$$+6\mu\{(A_-+A_+)(\Delta_-+\Delta_+)+\Delta_-\Delta_+\}+3(\sigma+3)A_-\theta_-+3(\sigma+1)A_+\theta_+. \tag{6.2}$$

PROOF. If zero lies in the support, all gaps have one sign and $J\le 2\int|t|\tau(t)\,dt=\rho$. Hence $D\ge\sigma\rho-\gamma\ge(\sigma-1)\rho$. Otherwise let $J_o=J-\mu(A_-+A_+)^2$ and

$$T_\pm=\mathbb{E}X_\pm^3-\mu A_\pm^2, \qquad R_\pm=\mathbb{E}[X_\pm(X_\pm-A_\pm)].$$

On a positive outer gap (a,b), $(b-a)^2\le 2\int_a^b(t-A_+)dt$. Summing and applying Tonelli, and then reflecting for the negative side, proves $J_o\le T_++T_--2A_+R_+-2A_-R_-$. Expansion gives the nonnegative decomposition

$$D=\mu(\sqrt{\sigma+1}A_--\sqrt{\sigma-1}A_+)^2+(\sigma-1)T_++(\sigma+1)T_-+3(T_++T_--J_o).$$

For the stronger assertion, conditional centering gives

$$\mathbb{E}X_\pm^3=\mu\bar A_\pm^2+\mathbb{E}[(X_\pm+2\bar A_\pm)(X_\pm-\bar A_\pm)^2;\ \pm X>0]\ge\mu\bar A_\pm^2+3A_\pm\theta_\pm,$$

and $R_\pm=\theta_\pm+\mu\Delta_\pm$. Insert these inequalities into the preceding outer-gap bound. Use $\mu^2=(1-\theta)pq$ and $(\sigma+1)p^2+(\sigma-1)q^2-6pq=(6+2\sigma)(p-p_*)^2$ to obtain (6.2). □

The upper bound for this same D comes from contact and Fourier inversion, rather than from an assumed two-point approximation. For $w=H^2/(2n)$ the exact conserved identity is

$$n(C-C_{\mathrm{E}})\rho+nC_{\mathrm{E}}\rho(1-e^{-w})+\frac{n\phi(0)e^{-w}}{6}D+n^{3/2}\mathcal{L}=\mathcal{R}, \tag{6.3}$$

where, setting

$$K_1=\mathbb{E}[X|X|^3]-3\beta-\tfrac{3}{2}\rho\gamma,\quad K_q=M_5/4-\rho M_4/2-\tfrac{3}{2}\beta\gamma+\rho/2,\quad U=J_1+(M_4-3)/6,$$

the right side is

$$\mathcal{R} = C(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(CK_1 + \varphi U) - H^2(C\rho + \gamma\varphi/2) + E_n. \tag{6.4}$$

The remainder has the exact representation

$$E_n = \frac{2\sqrt{n}}{\pi} \int_0^\infty \frac{\Im\{e^{-itH}[f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6)]\}}{t^3}\, dt, \qquad f(t) = \mathbb{E}e^{itX}. \tag{6.5}$$

This integral is absolutely convergent. To see the identity, write $\nu([x,\infty)) \le \mathfrak{q}(x)$ for the cubic majorant furnished by (5.2), with equality on the support. Evaluate $\int \tau(x)[\mathfrak{q}(x) - \nu([x,\infty))]dx$ first gap by gap and then using the density τ of the zero-bias law. The first evaluation is $\varphi J/(2\sqrt{n}) + G/n^{3/2} - \mathcal{L}$, where $G = 3C\beta J/2 + (H\varphi + 3C\rho)J_1 - 3CJ_2$. For the second, contact multiplied by X supplies the first truncated moment; the payoff $(x-z)|x-z|$ supplies the second truncated moment. Their difference gives

$$C\rho + \sqrt{n}\mathcal{L} = \frac{\varphi}{6}(\gamma + 3J) + (G - P_0)/n + E_n/n,$$
$$P_0 = H^2(C\rho + \gamma\varphi/2) - H\{CK_1 + \varphi(M_4 - 3)/6\} + CK_q.$$

Substitute $\gamma + 3J = (3+\sigma)\rho - D$ to obtain (6.3)–(6.4). Formula (6.5) follows from $\int_0^\infty (ty - \sin ty)t^{-3}dt = \pi y|y|/4$ by compensated Fubini; bounded support controls the origin and t^{-3} controls infinity. No discontinuous distribution function is inverted.

There is a necessary order in using this identity. A first bound $\mathcal{R} < r < nc_0$, with $C\rho \ge c_0 > 0$, proves $\varphi(\gamma + 3J)/6 = C\rho + \sqrt{n}\mathcal{L} - \mathcal{R}/n > 0$. Only then may the Gaussian weight be removed, giving

$$n(C - C_{\mathrm{E}})\rho + \kappa nD + n^{3/2}\mathcal{L} \le \mathcal{R}, \qquad \kappa = 997/15000 < \phi(0)/6. \tag{6.6}$$

Every later raw bound $\mathcal{R} \le r'$ also gives $e^{-w} \ge 1 - 5r'/(2n)$ and $nD \le r'/\kappa$. The initial pass therefore requires no defect bound of its own output. The actual central gap is used only after $D < (\sigma - 1)\rho$ has been established.

On a closed moment cell, the finite calculation bounds (6.5) by a coherent expression $K_F + s_F|H|$, and bounds the coefficients of (6.4). It thereby proves an envelope $\mathcal{R} \le \max_j(A_j + F_{j,\pm}|H| - Q_j^{\mathrm{raw}}H^2)$, where every $Q_j^{\mathrm{raw}} > 0$. Square completion bounds the envelope on the whole half-line. The signed curvature uses the actual γ: if $\gamma \ge g_0$ and $L_\phi \le \varphi \le U_\phi$, its valid price is $C\rho + \frac{1}{2}\min(L_\phi g_0, U_\phi g_0)$. Reflecting an auxiliary Fourier integral does not change this sign. Keeping the phase cost in (6.3) further gives

$$\kappa nD \le \max_j(A_j + F_{j,\pm}|H| - Q_jH^2). \tag{6.7}$$

For example, a previously proved $\mathcal{R} \le r$ and $D \le d$ allow the additional curvature $(1 - 5r/(2n))(.4095\rho - d/15)/2$, when positive. It is included once in Q_j. Thus a bound on the right side of (6.7) is converted to physical D using $15000/(997n)$ at the actual integer.

We now give the canonical parameters required by Section 2. Once the central gap is licensed, put

$$B = \mathbf{1}_{\{X>0\}}, \qquad \ell = \bar{A}_- + \bar{A}_+ = \mu/v, \qquad U = X/\ell + p - B. \tag{6.8}$$

Then $\mathbb{E}[U \mid B] = 0$ and $X = \ell(B + U - p)$ exactly. With $s_i = \mathbb{E}[U^2 \mid B = i]$, $s = qs_0 + ps_1$ and $\lambda_n = ns$,

$$\ell^2 = \frac{1-\theta}{v} = \frac{1}{v+s}, \qquad \mu\ell = 1 - \theta, \qquad \theta = \frac{s}{v+s}, \qquad \alpha = \frac{ps_1}{s} = \frac{\theta_+}{\theta} \quad (s > 0). \tag{6.9}$$

In particular, the moment in Section 2 is this law's actual ρ. If its physical support is in $[-R_-, R_+]$, take

$$a_0 = \Delta_-/\ell, \quad a_1 = \Delta_+/\ell, \quad O_0 = R_-/\ell - p, \quad O_1 = R_+/\ell - q. \tag{6.10}$$

These give precisely $-O_0 \le U_0 \le a_0$ and $-a_1 \le U_1 \le O_1$. The gap yields $a_0 + a_1 < 1$. Any zero outer cap can be enlarged to a positive cap when a comparison formula requires division by it. There is no probability symmetry in this construction: the original positive discrepancy determines the orientation of p.

For explicit resource coefficients, suppose the already acquired cell has $p \in I = [l, u] \subset (0, 1)$, $\theta \le \Theta < 1$, $A_- \ge A_l > 0$, $A_+ \ge B_l > 0$. Let

$$v_* = \max_{p \in I} p(1-p), \quad S_* = A_l + B_l,$$

$$b = \frac{(6 + 2\sigma)(1-\Theta)^{3/2}}{\sqrt{v_*}} \operatorname{dist}(p_*, I)^2, \quad g = 6(1-\Theta)S_*,$$

$$K_- = 3(\sigma+3)A_l, \quad K_+ = 3(\sigma+1)B_l, \quad h_{n,\pm} = \frac{K_\pm(1-\Theta)}{nv_*}.$$

Dropping the nonnegative product in (6.2) and using (6.9) proves the single inequality

$$D \ge b + g(a_0 + a_1) + \lambda_n\{(1-\alpha)h_{n,-} + \alpha h_{n,+}\}. \tag{6.11}$$

It supplies the upstream prices in (2.9), with $\mathfrak{R} = \overline{D} - b$ whenever $D \le \overline{D}$. Replacing $K_\pm$ by their minimum gives the scalar noise price used by the finite rows. Separate upper bounds for inward displacement and noise do not imply (6.11).

The stronger signed prices used in the new comparison are finite consequences of this inequality. Write $R = \overline{D} - b \ge 0$ and let $\mu_l > 0$ be an established lower bound for μ. The endpoint floors may first be improved by

$$A_l \leftarrow \max\left(A_l, \sqrt{\frac{(1-\Theta)l}{1-l}} - \frac{R}{6\mu_l S_*}\right),$$

$$B_l \leftarrow \max\left(B_l, \sqrt{\frac{(1-\Theta)(1-u)}{u}} - \frac{R}{6\mu_l S_*}\right),$$

using the old S_* in the denominator. Recompute $g, h_{n,\pm}$ from these proved floors. Combining (6.11) with $a_0 + a_1 \ge s_0/O_0 + s_1/O_1$ gives

$$\lambda_n\{(1-\alpha)C_0(p) + \alpha C_1(p)\} \le R, \quad C_0(p) = h_{n,-} + \frac{g}{n(1-p)O_0}, \quad C_1(p) = h_{n,+} + \frac{g}{npO_1}.$$

For fixed positive caps, C_0 increases and C_1 decreases in p. Thus the uniform allocation plane is

$$\lambda_n\{C_0(l) + [C_1(u) - C_0(l)]\alpha\} \le R. \tag{6.12}$$

When the separately checked slope $C_1(u) - C_0(l)$ is nonnegative, $\lambda_n \le R/C_0(l)$. More generally, positivity of both endpoint coefficients always gives

$$\lambda_n \le \frac{R}{\min\{C_0(l), C_1(u)\}}. \tag{6.13}$$

This consequence permits either ordering of the signed noise prices. It also contains the scalar harmonic bound. Indeed, for $B_* = \max(O_0, O_1)\max(u, 1-l)$ and $h_* = \min(h_{n,-}, h_{n,+})$,

$$\min\{C_0(l), C_1(u)\} \ge h_* + \frac{g}{nB_*}, \qquad \lambda_n \le \frac{R}{h_* + g/(nB_*)}. \tag{6.14}$$

This follows from $(1-l)O_0 \le B_*$ and $uO_1 \le B_*$. It improves θ through (6.9), and hence improves the next endpoint, outer-cap and price estimates. A prescribed finite number of these steps proves the strengthened rows by induction; each step uses the previous variance bound. It assumes neither a fixed point nor a special value of the unknown allocation α.

The saved floor-25 enclosures are interpreted as

$$\mathcal{R}_n \le r_{25}, \qquad nD \le 25D_{25}. \tag{6.15}$$

This is a fixed sequence of inequalities for the same actual-n law. Appendix A gives the complete low/high-frequency candidates, their index gates and the finite stage induction. The first unseeded raw bound licenses removal of the Gaussian weight before its defect output is used. Every subsequent step first transfers the incoming physical bounds, acquires the needed support at the actual integer, and then restores the weighted defect bound. Neither extremality at 25 nor monotonicity of an adaptively rounded program is assumed. Physical deductions may use the weaker $D \le D_{25}$.

For the exported plane set $r = 25/n$, $K = \min(K_-, K_+)$, $h_{25} = K(1-\Theta)/(25v_*)$ and $R = D_{25} - b$. If $R > 0$, write $d = R/g$ and $L = R/h_{25}$. Then the resulting scaled plane is

$$\frac{a_0 + a_1}{rd} + \frac{\lambda_n}{L} \le 1 - (n/25 - 1)\frac{b}{R} \le 1. \tag{6.16}$$

The joint noise intercept L stays fixed. Appendix A derives this plane, its zero-residual boundary and the weighted specialization of (6.14).

Finally, new physical support can improve these same parameters. From an established $\theta \le t < 1$, choose $0 < \ell_l \le \sqrt{(1-t)/v_*}$. Then

$$O_0 \le R_-/\ell_l - l, \qquad O_1 \le R_+/\ell_l - (1-u).$$

Intersect with previous caps and apply the same endpoint elimination; (6.9) then updates the variance and mean bounds. Each step consumes the preceding bound. A scalar consequence does not replace the joint intercept L in (6.16).

Consequently a finite acquisition must cover the entire initial moment and probability domain by closed cells; prove the support, Fourier and coefficient inequalities on each cell; and either exclude it by a strict contradiction or export the simultaneous quantities above. Its surviving rows are necessary conditions on the original global extremizer. For the surviving middle-band profiles, the acquired contact radii satisfy $H_-, H_+ < 2$, the original $\Lambda < 1/2$, and $p \in [.32, .43]$. Since $\lambda_n \le \Lambda$ and $n \ge 26$, $\sqrt{pq + \lambda_n/n} < 3/5$. Hence the raw threshold belongs to

$$np + H/\ell \in (np - 6/5, np + 6/5) \subset (0, n).$$

At this raw threshold the comparison uses $np + H/\ell = k + t$ with $|t| \le 1/2$, including both choices at a tie. Every nearest integer lies in $0 \le k \le n$; all such integers are covered or excluded by the acquired contact bounds. This supplies the continuous meaning of the finite comparison inputs, independently of their file labels.

7. The five finite-index steps. We now compose the comparisons at the actual integers $n = 26, \dots, 30$. The predecessor in each step is the unrestricted bound $C_{n-1} \le C_{\mathrm{E}}$. All parameters in a single step describe one extremizer supplied by Lemma 5.1. In particular, neither its third moment nor its noise is chosen independently for different count layers. Throughout this section $R(t)$ denotes (3.1) with the fixed target $c_n = C_{\mathrm{E}}/\sqrt{n}$. Computations can use a smaller directed rational coefficient: an upper bound for that larger residual is also an upper bound for this $R(t)$.

7.1. *Localization and the domain of the calculation.* The supporting localization proof gives three closed moment ranges. For $\rho \geq 6/5$, a support-free Prawitz comparison at each actual integer gives $R_n(P) < .4097 < C_{\mathrm{E}}$; the large-moment ray is included. For $1 \leq \rho \leq 11/10$, the contact and resource reductions, with the full predecessor bound, give complete conditional count covers. The original covers across the five integers consist of 79 probability intervals, 2289 count roots and 74170 terminal inequalities. Every other probability interval is excluded already during localization. These counts describe the coverage, not a replacement for the inequalities proving it. The present composition uses those ordinary count proofs on 72 intervals. Proposition 7.1 covers the remaining seven.

It remains to handle $11/10 \leq \rho \leq 6/5$. At each actual integer, localization covers the 100 original closed intervals $[i/100, (i+1)/100]$, $0 \leq i < 100$. It excludes 89 intervals and leaves exactly

$$I_i = [i/100, (i+1)/100], \qquad 32 \leq i \leq 42.$$

The four outer intervals $I_{32}, I_{33}, I_{41}, I_{42}$ have complete ordinary count covers. At $n = 26$, the same is true of I_{40}. The new comparisons therefore concern 34 parameter profiles: $I_{34}, \ldots, I_{39}$ at 26, and $I_{34}, \ldots, I_{40}$ at each of $27, 28, 29, 30$. Their scalar caps are listed in Appendix C.

A profile supplies inward and outward support caps Δ, O_0, O_1, a noise cap Λ, and the necessary joint inequalities

$$a_0 + a_1 \leq \Delta(1 - \lambda/\Lambda), \qquad g(a_0 + a_1) + \lambda\{(1-\alpha)\mathfrak{h}_- + \alpha\mathfrak{h}_+\} \leq \mathfrak{R}. \tag{7.1}$$

The second inequality is the consequence of (6.11) with $D \leq \overline{D}$ and $\mathfrak{R} = \overline{D} - b$. It can strengthen the first cap and supplies the signed plane (2.9). Its coefficients come from the actual-index resource proof. They are held fixed on their stated probability interval, not fitted to the count being estimated. The contact bounds also give

$$-H_- \sqrt{pq + \lambda/n} \leq k + t - np \leq H_+ \sqrt{pq + \lambda/n}, \qquad |t| \leq 1/2. \tag{7.2}$$

This leaves only three neighboring counts in each profile. The support caps, both resource bounds and the actual-moment floor are used simultaneously, with no finite-support assumption on the original conditional laws.

7.2. *Bounding a whole parameter region.* For a current-count price, let L_* be the acquired actual-noise cap. The seven original closed noise intervals have endpoints

$$0, \quad \Lambda/8, \quad \Lambda/4, \quad 5\Lambda/16, \quad 3\Lambda/8, \quad 7\Lambda/16, \quad \Lambda/2, \quad L_*;$$

the four allocation intervals have endpoints $0, 1/4, 1/2, 3/4, 1$. These endpoints are increasing in every profile. Their Cartesian product gives 28 fixed reference laws per current-count family. Each is valid for its entire closed region by Lemma 2.3; it is not a sample at its midpoint. The signed plane and the inward cap restrict the admissible conditional variances before the reference law is formed. The noise correction in Proposition 4.1 preserves the difference between that reference law and the actual noise. Probability faces and phase faces extend to the whole region by the corresponding lower-curvature bounds. A tilted phase face at the Gaussian inflection is used when needed.

For a reflected-square price, Proposition 4.2 keeps the actual noise and allocation until the two derivative gates have been proved. The resulting endpoint envelope uses the same moment floor as the original law. A positive-atom activation is covered by the weak derivatives (3.4). Complementary prices may then be combined through the finite lower envelope of Lemma 3.2, using their directed lower target coefficients. Only explicit reference prices are differentiated in these steps.

All parameter endpoints, chord coefficients and interval enclosures are rational. The square roots in the finite prices, and the normal CDF and density, are enclosed outwards. A stationary width is accepted only when it belongs to its asserted active set. Curvature bounds include activation boundaries. The finite maximum of every interpolating quadratic includes its interior vertex when present. Thus a negative upper bound applies to every point of the closed domain, including all shared faces. The full count roots and the unions of their closed phase intervals are retained when these regional bounds are composed.

7.3. *Examples of the finite comparisons.* An illustrative replacement occurs at $n = 26$, $p \in I_{35}$, count 9. The current-count price with the least legal slope is strict on $[-1/8, 9/32]$ except for the last phase strip of the first noise interval. On that first noise interval, $[0, \Lambda/8]$, retain the current price through $t = 1/4$ and use a coherent count-10 reflected-square price of width $9/8$ on $[1/4, 9/32]$. Its allocation and noise derivative margins are respectively at least $.120534962164$ and $.072056354791$, and its variance derivative is at least $.759364977755$. Its residual upper bound is $-.018464965470$ for its directed lower target coefficient. The other original noise intervals retain the current price on the whole strip. The resulting uniform upper bound on $[-1/8, 9/32]$ is

$$R(t) \le -.000067780727. \tag{7.3}$$

The adjacent negative phases have a full-square or current-count bound, and $[9/32, 1/2]$ has the full-square count-10 bound. Consequently this entire count layer uses only second-moment comparisons. The remaining count-10 regions in I_{36}, I_{37} likewise admit current-count comparisons with the rational slope multipliers $1, 5/4, 3/2$. No fourth-moment price or optimized deterministic-group slope is needed by these analytic profile comparisons.

A second completion illustrates why it is useful to distinguish the actual CDF from its price. At $n = 30$, $p \in I_{37}$, the count-10 contact interval begins at the exact outward bound

$$a_{37} = 249467754491/10^{12} < 1/4.$$

An endpoint estimate at $1/4$ extends over $[a_{37}, 1/4]$ by Lemma 3.3. The comparison used to prove that endpoint estimate need not remain valid on the short interval. Monotonicity of the original CDF is sufficient.

PROPOSITION 7.1 (Seven lower-moment profiles). *For each row below, retain the original actual-n support and resource acquisition and the unrestricted predecessor $C_{n-1} \le C_{\mathrm{E}}$. The whole profile $p \in I_i$, $1 \le \rho \le 11/10$ has conditional count residual bounded by the last column. The joint intercept Λ is retained; L_* is a necessary ceiling on the actual noise.*

n	I_i	Δ	Λ	L_*	*residual upper*
26	I_{39}	*.05769*	*.30708*	*.17375152*	$-.000098710877$
26	I_{40}	*.057809*	*.301161*	*.17805629*	$-.000005610547$
26	I_{41}	*.054236*	*.276951*	*.162619*	$-.000049410586$
27	I_{41}	*.052227*	*.276951*	*.16129388*	$-.000016889$
28	I_{41}	*.050362*	*.276951*	*.16009718*	$-.000027176644$
29	I_{41}	*.048625*	*.276951*	*.15901083*	$-.000059169468$
30	I_{41}	*.047004*	*.276951*	*.15802002*	$-.000017416596$

PROOF. Appendix D gives all seven finite plans and their closed count/phase unions. Each uses the seven original noise bands and four allocation quarters. The signed resource plane first gives L_*; current-count and reflected-square prices then cover every surviving contact interval, with the bounds in the table.

Three features require the full comparison rules: reversed signed prices use (6.13); the count-12 reference at $n = 29$ crosses np and uses both branches of (4.8); and the early $n = 30$ square has $y_{\max} > 0$, covered by the $y_{\max} < b$ bound in Section 3. The remaining steps use the same endpoint, curvature and threshold-shift identities, with each row's actual integer and acquired parameters. □

7.4. *The finite implications.* The complete middle-moment profiles at 27–30 give the following outward upper bounds on $R(t)$, uniformly over their probability interval, actual noise, allocation, all surviving counts and contact phases. For readability, the table reports 10^6 times these bounds, rounded upwards to three decimal places.

interval	$n = 27$	$n = 28$	$n = 29$	$n = 30$
I_{34}	-320.741	-263.994	-321.528	-241.967
I_{35}	-297.200	-134.578	-287.407	-436.298
I_{36}	-294.861	-12.020	-96.700	-156.955
I_{37}	-145.182	-12.884	-1.810	-63.857
I_{38}	-70.159	-56.159	-79.848	-22.937
I_{39}	-35.366	-4.435	-585.317	-75.052
I_{40}	-1160.650	-723.488	-713.730	-987.882

The six analytic profiles at 26 are also strictly negative; (7.3) describes its only former fourth-moment region. The full results include the exact resource caps, count domains, price choices, derivative gates and outward rational bounds, rather than only these decimal margins.

PROPOSITION 7.2 (The five actual-index implications). *For each* $n = 26, 27, 28, 29, 30$*,*

$$C_{n-1} \le C_{\mathrm{E}} \quad \Longrightarrow \quad C_n \le C_{\mathrm{E}}.$$

PROOF. If $C_n > C_{\mathrm{E}}$, Lemma 5.1 supplies a positive closed extremizer. The full predecessor bound makes its allowance (5.4) zero, so the actual-index localization and resource inequalities apply to this same law. The support-free Fourier estimate excludes $\rho \ge 6/5$. The complete lower-moment count covers exclude $1 \le \rho \le 11/10$. In the remaining closed band, localization either gives an immediate contradiction or places the law in one of the eleven intervals above. The ordinary count covers and the 34 analytic profiles exclude them all. Every surviving count and its entire contact interval has a strict negative bound, contradicting the positive extremal score. The three closed moment ranges exhaust $\rho \ge 1$. □

THEOREM 7.3 (The iid Esseen-constant inequality). *For every standardized real distribution* P *with* $\rho(P) = \mathbb{E}|X|^3 < \infty$ *and every positive integer* n*,*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3 + \sqrt{10}}{6\sqrt{2\pi}}\,\rho(P).$$

The constant is optimal over this class and all n*.*

PROOF. Input (I) of Proposition 5.3 supplies the unrestricted bound at each index through 25. Apply Proposition 7.2 in order, discharging each global predecessor, to obtain the bounds through 30. In particular $C_{30} < C_{\mathrm{E}} + 1/100$, so input (II) gives 31–46. Proposition 5.4 then gives every $n \ge 47$ from its two sharp predecessors and the uniform local exclusion (III). These four ranges exhaust all positive integers.

For the standardized Bernoulli law with $p_* = (4-\sqrt{10})/2$, the classical binomial asymptotic gives

$$\lim_{n\to\infty} \frac{\sqrt{n}\,\Delta_n(P_*)}{\rho(P_*)} = C_{\mathrm{E}};$$

see Schulz [3, Remarks 4.1 and 4.2]. Thus a smaller uniform constant is impossible. □

8. Scope of the supporting material. The finite comparisons in Sections 2–4 are analytic inequalities for arbitrary real conditional laws. The finite calculations certify their use on the closed parameter domains of Section 7. The supporting material also contains the continuous localization and the proofs of inputs (I)–(III). These are essential parts of the all-index argument, not hypotheses inferred from the numerical table.

The moment-defect and localization framework, the original subdivisions, and the imported count and Fourier arguments are due to Rodziewicz and Kołodziejek [8]. The role of the present comparison method is to close the remaining general-law regions at the five actual integers and compose the supporting implications in a well-founded order.

This working version includes the companion mathematical sources, exact acquired parameters, collected frozen arithmetic inputs and archived calculation outputs. The accompanying programs locate these materials and run their mathematical readers from the collected directories. Fresh numerical regeneration uses the retained producer entrypoints and their stated runtime requirements. The collection is local at this stage; no permanent public archive identifier is assigned in this draft.

AI assistance. OpenAI Codex was used to develop and simplify mathematical arguments, implement exact-arithmetic calculations, and draft this manuscript.

APPENDIX A: A FIXED-FLOOR FOURIER AND RESOURCE TRANSFER

This appendix proves the interpretation of the fixed floor-25 acquisition used in (6.15). The floor is an evaluation parameter in a collection of inequalities. It is not another sample size at which the current law is assumed to be extremal. The moment-defect identity and the underlying enclosure method retain their attribution to Rodziewicz–Kołodziejek [8].

Fix $N = 25$, an integer $n \in \{26, \dots, 30\}$, and a standardized law P attaining the unrestricted $C_n = C \geq C_{\mathrm{E}}$ at a positive closed threshold H. Assume $C_{n-1} \leq C_{\mathrm{E}}$ and $\rho \in [1, 11/10]$ or $\rho \in [11/10, 6/5]$. Use D, $\mathcal{L}$ and $\mathcal{R}_n$ from (6.3)–(6.4), all for this actual pair (P, H). Write $\mathcal{R}_n = \mathcal{R}$ for the right side of the conserved identity (6.3) at this same law. We require an actual-index score cap $C \leq c_{\mathrm{in}}$ with $c_{\mathrm{in}} \leq 434477/10^6$; the Gaussian replacement bound $C < .422$ under the sharp predecessor is sufficient. The support and moments used below must have been acquired for this same actual-n pair. No assertion about $C_{25}(P)$ is required.

A.1. The complete low-frequency estimate. Put $f(t) = \mathbb{E}e^{itX}$, $\gamma = \mathbb{E}X^3$, and

$$V_n(t) = f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6),$$

$$I_0(n, H) = \int_0^{1/2} \frac{|\Im(e^{-itH} V_n(t))|}{t^3}\, dt,$$

$$I_\infty(n) = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

By (6.5),

$$|E_n| \le \frac{2}{\pi}\{\sqrt{n} I_0(n,H) + \sqrt{n} I_\infty(n)\}. \tag{A.1}$$

Both integrals are retained, including the entire Gaussian comparator on $[1/2,\infty)$.

LEMMA A.1 (Low-frequency candidates). *Suppose $n > 2$, $M_4 \le M \le 111/10$, $|\gamma| \le g \le 37/20$, and $M_6 \le \overline{M}_6$. For the actual coefficients*

$$a = (3 - M_4)/24, \qquad z_5 = (10\gamma - \mathbb{E}X^5)/120, \qquad c_5 = a\gamma/6,$$

let $a_* \ge |a|$ and $Z \ge |z_5|$. Define, for real $j > 2$,

$$A_j = \frac{j-1}{2j}, \quad B_j = \frac{45 - M}{96}(1 - 2/j), \quad F_j = (j/(j-1))^{3/2}, \quad I_u \ge \sqrt{\pi/2},$$

$$E_6 = \frac{\overline{M}_6}{720} + \frac{g^2}{72} + \frac{(1/2 + g^2/144)^3}{5}, \quad T_6 = a_* + Z/2 + E_6/4, \quad L_2(j) = \frac{E_6}{2A_j^2} + \frac{T_6^2}{2B_j^3}.$$

Each of the following complete triples satisfies

$$\sqrt{n} I_0(n,H) \le L_0(n) + L_H(n)|H| + L_2(n)/\sqrt{n}. \tag{A.2}$$

The triangle triple is

$$L_0(j) = I_u F_j (Z + a_* g/2) + 251 g^3/17280,$$

$$L_H(j) = I_u F_j a_* + 251 g^2/4800. \tag{A.3}$$

For the cancellation triple, let a closed rectangle contain the actual (z_5, c_5), and let $J_* \ge 0$ bound

$$J(z,c) = \int_0^\infty e^{-x^2/2} |zx^2 - cx^4|\, dx$$

at every corner of that rectangle. Then one may take

$$L_0(j) = F_j J_* + \frac{F_j I_u}{j-1}(35 a_* g^3/48 + 35 Z g^2/24) + 251 g^3/17280,$$

$$L_H(j) = F_j I_u \{a_* + 5Zg/[2(j-1)]\} + 251 g^2/4800. \tag{A.4}$$

In either case, all three coefficients and $L_2(j)/\sqrt{j}$ are nonincreasing for $j \ge N$ when their physical inputs are fixed.

PROOF. Set $h(t) = e^{-t^2/2 - i\gamma t^3/6}$. Taylor expansion gives

$$f(t) - h(t) = -at^4 - iz_5 t^5 + r_6(t), \qquad |r_6(t)| \le E_6 t^6 \quad (0 \le t \le 1/2).$$

Indeed, for $u = -t^2/2 - i\gamma t^3/6$, $|u| \le t^2(1/2 + g^2/144) < 1/6$; the omitted exponential terms cost at most $|u|^3/5$, and its degree-six quadratic term costs $g^2 t^6/72$. Symmetrization and $\cos y \le 1 - y^2/2 + y^4/24$ give

$$|f(t)|, |h(t)| \le e^{-(45-M)t^2/96}.$$

Taylor expansion of the polynomial w^n along the joining segment therefore yields

$$f(t)^n - h(t)^n = n(f(t) - h(t))h(t)^{n-1} + R_2(t), \qquad |R_2(t)| \le \tfrac{1}{2} n^2 T_6^2 t^8 e^{-nB_n t^2}.$$

Integrating $n r_6 h^{n-1}$ and R_2, after division by t^3, costs at most $L_2(n)/n$. For $u = n\gamma t^3/6$, rotation by Ht bounds the remaining Gaussian comparator's imaginary part

by $e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2)$. The Gaussian even moments give its integral at most $n^{-1/2}(251g^3/17280 + 251g^2|H|/4800)$.

For the leading quartic and fifth terms put $b = n - 1$ and $v = b\gamma t^3/6$. The bounds $|\sin(Ht + v)| \le |H|t + bgt^3/6$ and $|\cos| \le 1$ give (A.3). Alternatively, the leading imaginary part after division by t^3 is bounded, before multiplication by $ne^{-bt^2/2}$, by

$$|z_5 t^2 - atv| + a_* t|\sin v - v| + Zt^2|\cos v - 1| + |H|t\{a_* t + Zt^2|\sin v|\}.$$

The first integral is $nJ(z_5, c_5)/b^{3/2}$. The norm J is convex in (z, c), so its rectangle maximum is bounded by the largest of its corner bounds. Integrating the other four terms with the global sine and cosine remainders gives (A.4). These arguments use the original H, without rescaling it to a floor-N threshold.

Finally,

$$\frac{F_j'}{F_j} = -\frac{3}{2j(j-1)}, \quad A_j' = \frac{1}{2j^2} > 0, \quad B_j' = \frac{45 - M}{48j^2} > 0.$$

Thus F_j, $F_j/(j-1)$, A_j^{-2}, B_j^{-3} and $j^{-1/2}$ decrease. Every multiplier of these factors in (A.3)–(A.4) is nonnegative, including J_*. This proves each claimed coefficient comparison. □

Here are the coefficient choices actually needed in these candidates. They are physical moment inequalities, independent of the sample size. For the triangle choice, assume $|X| \le R \le 6$ and $M_5 \le \eta$. One may use $a_* = \max(2, M - 3)/24$, $\overline{M}_6 = R\eta$, and $120Z = \min\{\eta + 10g, Z_6, Z_r\}$, where

$$Z_6 = \begin{cases} 6(M-1) + (10 - 2M)g - 6g^2 + g^3, & M \le 5,\ 10 - 2M - 12g + 3g^2 \ge 0, \\ 6(M-1) + (10 - 2M)^2/[4(6-g)], & M \le 5 \text{ otherwise}, \\ 6(M-1) + (2M - 10)^2/24, & M > 5, \end{cases}$$

and, with $r_0 = \max(R, 2g) > g$, put

$$t_* = \begin{cases} \min(g, (10 - 2M)/(2(r_0 - g))), & M \le 5, \\ \min(g, (2M - 10)/(2r_0)), & M > 5. \end{cases}$$

Then

$$Z_r = \begin{cases} r_0(M-1) + (10 - 2M)t_* - (r_0 - g)t_*^2, & M \le 5, \\ r_0(M-1) + (2M - 10)t_* - r_0 t_*^2, & M > 5. \end{cases}$$

To verify them, reflect only the auxiliary Fourier integral so that $t = \gamma = |\gamma| \ge 0$, and set $Q = X^2 - tX - 1$. Direct multiplication gives

$$\begin{aligned} \mathbb{E}Q^2 &= M_4 - 1 - t^2, \\ \mathbb{E}X^5 &= t^3 + 2t + \mathbb{E}[Q^2(X + 2t)], \\ M_6 &= 1 + 3t^2 + t^4 + \mathbb{E}[Q^2\{(X + t)^2 + 2t^2 + 2\}]. \end{aligned} \tag{A.5}$$

For an enclosing radius $r \ge 2g$, the two signs of $\mathbb{E}X^5 - 10t$ are bounded by $r(M-1) \pm (10 - 2M)t - rt^2 \pm t^3$. Use $t^3 \le gt^2$, or discard the negative cubic, and maximize the resulting concave quadratic on $[0, g]$. At $r = 6$, the increasing endpoint alternative gives the first line of Z_6.

The resource implementation also uses the following general signed rectangle, which does not require $M < 3$ or $R > 2g$. Suppose $0 \le g_l \le |\gamma| \le g$, $M \ge 1 + g_l^2$, and $R \ge 1$. From (A.5), for every feasible $t \in [g_l, g]$,

$$L(t) \le 120 z_5 \le U(t),$$

$$L(t) = -R(M-1) + (10 - 2M)t + Rt^2 + t^3,$$

$$U(t) = \begin{cases} R(M-1) + (10-2M)t - Rt^2 + t^3, & t \le R/2, \\ 8t - t^3, & t \ge R/2. \end{cases}$$

The second line for U uses $\mathbb{E}Q^2 \ge 0$ when $R - 2t < 0$; using $M - 1 - t^2$ there would have the wrong direction. Choose outward bounds $L_* \le \min L$ and $U_* \ge \max U$ on the indicated closed intervals, and set

$$z_l = \max(-\eta - 10g, L_*)/120, \quad z_u = \min(\eta + 10g, U_*)/120, \quad Z = \max(|z_l|, |z_u|),$$

$$a_* = \max(|3 - M|, |2 - g_l^2|)/24,$$

$$c_l = \frac{3 - M}{144} \begin{cases} g_l, & M \le 3, \\ g, & M > 3, \end{cases} \qquad c_u = \max_{g_l \le t \le g} (2t - t^3)/144,$$

$$\overline{M}_6 = \min\{R\eta, 1 + 3g^2 + g^4 + (R^2 + 2Rg + 3g^2 + 2)(M - 1 - g_l^2)\}.$$

Check $z_l \le z_u$, $c_l \le c_u$ before using the rectangle. The cubic extrema can be enclosed by the Bernstein identity on a finite closed subdivision; this is an enclosure of the entire interval, rather than evaluation at finitely many sample points. The displayed a_* uses $1 + g_l^2 \le M_4 \le M$. The last line follows from the final identity in (A.5), with all substituted factors nonnegative. These prove the hypotheses of Lemma A.1 for the general signed candidate used by both resource branches. The restricted $M < 3$, $R > 2g$ formula is not needed to license a call outside its own domain.

The resource branch enables this signed candidate only on an already acquired interval $0 < l \le p \le u < 1/2$, with $\theta \le \Theta < 1$ and $A_- \ge A_l > 0$, $A_+ \ge B_l > 0$. Write $\rho_B(p) = \{1 - 2p(1-p)\}/\sqrt{p(1-p)}$ and set

$$\rho_* = \max\left\{1, \rho_l, \rho_B(u) - \max(0, 3\rho_B(l)/2 - 3\min(A_l, B_l))\Theta\right\},$$

$$g_l = \max\left\{0, \sigma_l \rho_* - 3 - d_D, \frac{(1-\Theta)^{3/2}(1-2u)}{\sqrt{u(1-u)}} - (R + 2\sqrt{u/(1-u)})\Theta\right\}.$$

Here $D \le d_D$ is an incoming bound. These are genuine physical lower bounds for ρ and $|\gamma|$, respectively. For the first, conditional expansion gives $\rho \ge (1-\theta)^{3/2}\rho_B(p) + 3\min(A_l, B_l)\theta$; use $(1-\theta)^{3/2} \ge 1 - 3\theta/2$ and the monotonicity of ρ_B on $(0, 1/2)$. For the second, $\gamma \ge \sqrt{10}\rho - 3 - D$ follows from $J \le \rho + 1$, whereas conditional expansion about the two means gives

$$\gamma \ge \frac{(1-\Theta)^{3/2}(1-2u)}{\sqrt{u(1-u)}} - (R + 2\sqrt{u/(1-u)})\Theta.$$

In the latter calculation the positive-class residual is nonnegative and the negative-class residual has magnitude at most the displayed coefficient times θ. The gates are $\rho_* \le \rho_u$, $0 \le g_l \le g \le 37/20$, $1 \le M \le 111/10$, $R \ge 1$ and $M - 1 - g_l^2 \ge 0$. The same ρ_* is kept in the paired rectangle, score corners, phase callback and exports. On another probability half, or if this candidate is ineligible, the unpaired candidate remains available; ineligibility alone does not exclude a law.

For completeness, the smaller Gaussian corner prices used in this rectangle also have an index-free proof. If $zc \le 0$, then $J(z,c) = \sqrt{\pi/2}(|z| + 3|c|)$. If $zc > 0$, replace (z,c) simultaneously by $(|z|,|c|)$ and put $x = z/c > 0$. Splitting the integral at $\sqrt{x}$ gives

$$J(z,c) = \sqrt{\pi/2}(z - 3c) + \sqrt{zc}e^{-x/2}\left(6 + \frac{2(3-x)}{x}\sqrt{x}\,\mathcal{M}(\sqrt{x})\right),$$
$$\mathcal{M}(r) = \int_0^\infty e^{-rt-t^2/2}dt. \tag{A.6}$$

For $I_j(r) = \int_0^\infty t^j e^{-rt-t^2/2}dt$, integration by parts gives $I_0 = 1/(r + I_1/I_0)$ and $I_j/I_{j-1} = j/(r + I_{j+1}/I_j)$. Truncating this finite recursion with a zero last ratio gives an alternating upper bound at even depth and lower bound at odd depth. In rational form, set $d_{k+1} = 1$, $d_j = 1 + j/(xd_{j+1})$ for $j = k, \dots, 1$. In (A.6) replace $\sqrt{x}\mathcal{M}(\sqrt{x})$ by d_1^{-1} with $k = 11$ if $x \ge 3$, and $k = 12$ if $x < 3$. The sign of $3 - x$ gives the required upper bound in both cases. The remaining multiplier is positive; for $x \ge 3$ it exceeds $6 - 2(x-3)/x > 4$. A negative coefficient $z - 3c$ uses a lower enclosure of $\sqrt{\pi/2}$ and a positive coefficient uses an upper enclosure. The minimum with

$$\sqrt{\pi/2}|z - 3c| + 6\sqrt{zc}e^{-x/2}$$

is also a valid corner bound. These are the two corner alternatives in the saved candidates. Auxiliary reflection here leaves the original signed γ in the contact identity unchanged.

A.2. The complete high-frequency estimate. Let X' be an independent copy of this same X. A candidate consists of fixed numbers M, a, m, d, w and an actual event E such that

$$M_4 \le M \le 111/10, \qquad \mathbb{P}(E) \ge a > 0,$$
$$|X - X'| \ge d > 0 \text{ on } E, \qquad \mathbb{E}[(X - X')^2 \mid E] \le m, \tag{A.7}$$
$$0 < d \le 12, \qquad 0 < w \le 1/4.$$

Here $a \le 1$ and $m > 0$. Put $b = 9/10$,

$$h = 333/106 - (22/7)^3 w^2/6 > 0, \quad \kappa_E = a(23/48)^2 d^2, \quad \Lambda(x) = \frac{x+2}{x^2+4x+2},$$

$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20}(j-w)^{-5} + [4(20-w)^4]^{-1},$$

$$c = \tfrac{1}{2}\left(1 - \max_{u \in \{1/4, 81/100\}}[1 - u + (M+3)u^2/12]\right),$$

$$c_0 = 1/8 - (M+3)/384, \qquad s = \{1 - (M+3)53/600\}/2.$$

Define

$$\mathfrak{H}_j = A_j^{\text{high}} + \frac{m}{4\pi^{3/2}h\sqrt{a}}\left(S_3 + \frac{3S_5}{jah^2}\right) + \frac{m\sqrt{j}}{2\pi^2}e^{-jah^2w^2}$$
$$+ Z_j + \frac{\sqrt{j}}{2b^2}e^{-ja(23/48)^2} + \sqrt{j}e^{-j/8}\{2\Lambda(j/8) + \sqrt{M-1}/3\}. \tag{A.8}$$

One always may choose

$$A_j^{\text{high}} = 2\sqrt{j}e^{-jc}, \qquad Z_j = \frac{e^{-j\kappa_E b^2}}{2b^4\kappa_E\sqrt{j}}.$$

If $s > 0$, the first may be replaced by its minimum with $2\sqrt{j}e^{-jc_0}\Lambda(js/4)$. If $j\kappa_E b^2 \ge 1/4$, the second may be replaced by its minimum with $\sqrt{j}e^{-j\kappa_E b^2}\Lambda(j\kappa_E b^2)/(2b^2)$.

LEMMA A.2 (High-frequency candidates and their gates). *Under* (A.7), $\sqrt{n}I_\infty(n) \le \mathfrak{H}_n$. *A fixed choice of alternatives satisfies* $\mathfrak{H}_n \le \mathfrak{H}_N$ *for every* $n \ge N$ *provided*

$$2Nc' > 1 \quad (c' \in \{c, ah^2w^2, a(23/48)^2, 1/8\}). \tag{A.9}$$

For the improved first term also require $s > 0$ *and* $2Nc_0 > 1$*; for the improved zero-cell term require* $N\kappa_E b^2 \ge 1/4$*. An alternative failing its floor gate is omitted, even if it would become eligible at a larger index.*

PROOF. On $[1/2, b]$, symmetrization gives $|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12$, proving the first term with rate c. The sharper chord is

$$t^2/2 - (M+3)t^4/24 \ge c_0 + s(t^2 - 1/4).$$

It is valid since the difference is $(M+3)(t^2 - 1/4)(81/100 - t^2)/24 \ge 0$.

For $t \ge b$, let $p_E = \mathbb{P}(E)$. Symmetrization, $1 - u \le e^{-u}$, and conditional Jensen imply

$$|f(t)|^n \le e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-na\sin^2(t|X-X'|/2)} \mid E].$$

For each conditional distance $D_* = |X - X'|$, partition the positive axis into the zero cell and the cells centred at $2\pi j/D_*$, $j \ge 1$. In a nonzero cell write $t = (2\pi/D_*)(j+v)$. On $|v| \le w$, $|\sin(\pi v)| \ge h|v|$; on the wings it is at least hw. The paired Taylor bound

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-w)^{-5}v^2$$

and the Gaussian zeroth and second moments prove the second term of (A.8). Its Jacobian is $D_*^2/(4\pi^2)$, and conditional averaging uses $\mathbb{E}[D_*^2 \mid E] \le m$. Summing the two positive series and bounding their tails after 20 by integrals gives S_3, S_5. The wing weight is at most $D_*^2/(2\pi^2)$ and gives the third term. All extensions of these integrals below b add nonnegative quantities.

In the zero cell, $u = tD_*/2 \le \pi/2$. For $u \le 1/2$ use $\sin u \ge (23/24)u$ and $D_* \ge d$. This gives the zero-cell Gaussian tail Z_j. For $u \ge 1/2$ use $\sin u \ge 23/48$ and total weight $1/(2b^2)$, giving the fifth term. The comparator on the whole interval $[1/2, \infty)$ costs

$$\sqrt{j}\int_{1/2}^{\infty} e^{-jt^2/2}t^{-3}dt + \frac{j^{3/2}|\gamma|}{6}\int_{1/2}^{\infty} e^{-jt^2/2}dt.$$

The second integral is at most $2e^{-j/8}/j$, and Pearson gives $|\gamma| \le \sqrt{M-1}$. This proves the sixth term.

The rational Gaussian-tail estimate used above is, for $x = jkb^2 > 0$,

$$\sqrt{j}\int_b^{\infty} e^{-jkt^2}t^{-3}dt \le \frac{\sqrt{j}e^{-x}}{2b^2}\Lambda(x). \tag{A.10}$$

After substitution its remaining factor is $\int_0^\infty u(x+u)^{-1}e^{-u}du$. For $P(u) = u^2 - 4u + 2$ and $D_x = x^2 + 4x + 2 = P(-x)$, division of $D_x^2 - P(u)^2$ by $x+u$, followed by $\int u^k e^{-u}du = k!$, gives $\int_0^\infty (x+u)^{-1}e^{-u}du \ge (x+3)/D_x$. The discarded remainder is $P(u)^2/[(x+u)D_x^2] \ge 0$. Consequently the preceding factor is at most $1 - x(x+3)/D_x = \Lambda(x)$, proving (A.10). The simpler zero-cell alternative follows by replacing t^{-4} by b^{-4} after integrating once.

For the floor comparison, $(\sqrt{j}e^{-cj})' = (\sqrt{j}e^{-cj})(1/(2j) - c)$, and $j^{-1/2}e^{-cj}$ decreases for every $c > 0$. Also $\Lambda'(x) = -(x^2+4x+6)/(x^2+4x+2)^2 < 0$. These prove the assertions for the second, third, fifth and sixth terms and for either first-term choice under the stated gates. For the improved zero-cell choice the derivative of $\sqrt{x}e^{-x}\Lambda(x)$ has, after multiplication by positive factors, numerator

$$4 - 10x - 22x^2 - 13x^3 - 2x^4.$$

It equals $-11/128$ at $x = 1/4$ and decreases for $x > 0$. This proves precisely its additional gate. □

The event inputs transfer without an index assumption. For $0 < d^2 < 2$ and $E = \{(X - X')^2 \geq d^2\}$, Cantelli's inequality supplies

$$V = 2M + 2, \quad h_0 = 2 - d^2 > 0, \quad a = h_0^2/(V + h_0^2), \quad m = 2 + V/h_0.$$

Indeed $Y = (X - X')^2$ has $\mathbb{E}Y = 2$ and $\operatorname{Var} Y \leq V$. Applying the elementary one-sided second-moment bound to $Y - 2$ gives $\mathbb{P}(Y \geq d^2) \geq a$. For $p_E = \mathbb{P}(E)$, Cauchy–Schwarz applied to the covariance gives $\mathbb{E}[Y \mid E] \leq 2 + \sqrt{V(1 - p_E)/p_E} \leq 2 + V/h_0$. This also licenses the older weaker parameters $a_U = h_0^2/(V + 4)$, $m_U = 2(V + 4)/h_0^2$: $a \geq a_U$ and

$$m_U - m = \frac{(2 - h_0)(V + 2h_0 + 4)}{h_0^2} \geq 0.$$

Every event-dependent term of (A.8) decreases in a and increases in m; its gates are preserved when a increases. Thus no separate universal-event proof is needed, and old certificates using the weaker parameters remain valid. This domination holds for the same M, d, w; it does not identify different distance choices. The selected Cantelli pool uses $d \in \{1/2, 11/20, 3/5, 13/20, 7/10\}$ and $w = 1/4$. Only members satisfying (A.9) at N are retained. Replacing M by a fixed outward ceiling on the $1/1000$ grid is legitimate as a new fourth-moment cap; it is then held fixed.

After a central gap $(-A_-, A_+)$ is proved, another actual event is $E = \{XX' < 0\}$. With $p = \mathbb{P}(X > 0)$, $\mu = \mathbb{E}X_+$ and $\beta = \mathbb{E}[X|X|]$, its exact conditional distance is

$$\mathbb{P}(E) = 2p(1 - p), \qquad \mathbb{E}[(X - X')^2 \mid E] = \frac{1 + 4\mu^2 + \beta}{2p} + \frac{1 + 4\mu^2 - \beta}{2(1 - p)}.$$

The expression increases in μ^2, is affine in β, and is convex in p for $|\beta| \leq 1$. Incoming interval endpoints therefore give m; the independent bound $m \leq 2/a$ may be intersected. An acquired $D \leq d_D < (\sigma_l - 1)\rho_l$, where $1 < \sigma_l \leq \sqrt{10}$, licenses the central gap. Moreover

$$(A_- + A_+)^2 \geq \frac{\rho_l - d_D/(\sigma_l - 1)}{\mu}, \qquad (A_- + A_+)^2 \geq \frac{2P_*}{3\mu}, \tag{A.11}$$
$$P_* = \max\{(\sigma_l - 1)\rho_l, \sigma_l\rho_l - \gamma_u\} - d_D.$$

The first follows from the gap decomposition's $T_+ + T_- \leq D/(\sqrt{10} - 1)$. The second follows from $J \leq \rho + 2\mu A_- A_+$ and hence $6\mu A_- A_+ \geq P_*$. Use an incoming upper bound for μ and positive lower roots. The weaker first bound with $\mu \leq 1/2$, and independently acquired $A_- + A_+ \geq A_l + B_l$, give the other saved distance choices. The gates are a positive gap, positive mean denominator and $0 < d \leq 12$; no assumption $d_D < 2$ is needed for (A.11).

A finite minimum of complete high-frequency candidates is valid: each bounds the same $\sqrt{n} I_\infty(n)$. Likewise a finite minimum of complete low-frequency triples is valid. One may not combine the constant of one low triple with the slope of another without a separate proof. Rational outward evaluations at N bound the exact displayed expressions at N and hence at n; no monotonicity of the rounding or exponential evaluator is assumed.

A.3. Raw and charged candidates at the fixed floor. Choose one complete low triple and one complete high bound at N. For a fixed rational $\omega_\pi \geq 2/\pi$, put

$$K_F = \omega_\pi\{L_0(N) + \mathfrak{H}_N + L_2(N)/\sqrt{N}\}, \qquad s_F = \omega_\pi L_H(N). \tag{A.12}$$

Lemmas A.1 and A.2 prove $|E_n| \leq K_F + s_F|H|$ for the actual law. The values $16/25$ and $212/333$ used in the two enclosure families both satisfy the normalization condition.

Here is a precise description of the coefficient estimates which can be used with (A.12). On an incoming physical cell assume that the coefficient atlas has proved

$$3\beta J/2 + 3\rho J_1 - 3J_2 - K_q \leq A(\rho), \qquad K_1 \in [k_l, k_u], \quad U \in [u_l, u_u]. \tag{A.13}$$

These are inequalities in the moments and support of P; they contain no sample index. For the initial cover they are proved on the signed $(\rho, \beta, \gamma, \mu)$ rectangles. For the resource cover they use the conditional moment and gap formulas of Section 6. The full physical derivations are the sections *Signed moments with a common first moment*, *Preserving central and outer integral costs* and *Moment and gap coefficients for a whole cell* in the continuous companion's *Coefficient atlas for the enclosure*, located in Appendix E. The central polynomial is $\rho P_1(A_-, A_+) - P_2(A_-, A_+)$, where

$$P_1 = B^3/3 + AB^2/2 - A^3/6,$$

$$P_2 = B^4/4 + AB^3/3 - A^4/12,$$

where $(A, B) = (A_-, A_+)$. Its full-rectangle Bernstein bound is a maximum of affine functions of ρ. The outer cost is bounded by $3\{(\rho - B_l)_+^2 T_+ + (R - \rho)^2 T_-\}/8$. These and the remaining affine moment terms give the convex $A(\rho)$ used in the corner reduction. Every signed product with a possibly negative factor retains the appropriate two endpoints. Thus (A.13), rather than a file status, is the required physical-coefficient premise of a candidate.

Suppose previous information gives $\mathcal{R}_n \le r$, $nD \le Nd$ and $0 < r < 2N/5$. Once removal of the Gaussian weight has been licensed, it follows that

$$D \le d, \qquad L_N \le \varphi \le 2/5, \quad L_N = \frac{997}{2500}(1 - 5r/(2N)) > 0. \tag{A.14}$$

The density statement uses the nonnegative phase term in (6.3), $C_{\mathrm{E}}\rho > 2/5$ and $n \ge N$. For the unseeded first pass use $L_N = 0$, without assuming r or d.

For a sign $\varepsilon \in \{-1, 1\}$ and a prospective score c, set

$$F_\varepsilon(c) = s_F + \max_{k \in \{k_l, k_u\},\, u \in \{u_l, u_u\},\, phi \in \{L_N, 2/5\}} \varepsilon(ck + \phi u).$$

The maximum of the two signs may be used for an unsigned bound. If $\gamma \ge g_0(\rho)$ is an incoming signed lower bound, with g_0 affine in ρ (including a fixed signed interval endpoint), the correct phase curvature is

$$q(c, \rho) = c\rho + \tfrac{1}{2} \min\{L_N g_0(\rho), (2/5) g_0(\rho)\}. \tag{A.15}$$

In particular $g_0(\rho) = 3.16\rho - 3 - d$ is valid since $J \le \rho + 1$. The alternative general-envelope curvature is

$$q_{\mathrm{gen}}(c, \rho) = c\rho + L_N g_0(\rho)/2 - (2/5 - L_N)(d - .16)_+/2.$$

It is no larger than (A.15), because $(-g_0(\rho))_+ \le (d - .16)_+$ for $\rho \ge 1$. Either curvature must be proved strictly positive before division. Using $L_N g_0/2$ alone when $g_0 < 0$ would not be licensed.

For the actual H with sign ε, (6.4) now gives

$$\mathcal{R}_n \le K_F + CA(\rho) + F_\varepsilon(C)|H| - q(C, \rho)H^2. \tag{A.16}$$

Thus a floor candidate is an envelope for the actual raw threshold. It is not an estimate of a fictitious $\mathcal{R}_{25}(P)$.

The same incoming bounds give a conserved-phase price

$$e_N(\rho) = \tfrac{1}{2}(1 - 5r/(2N))(.4095\rho - d/15) > 0. \tag{A.17}$$

Indeed the difference between the weighted and unweighted terms in (6.3) is $n(1 - e^{-w})(C_{\mathrm{E}}\rho - \phi(0)D/6)$, $w = H^2/(2n)$. Use $1 - e^{-w} \ge we^{-w}$, the density floor before multiplying by $\phi(0)$, $D \le d$ and $\phi(0) < 2/5$. Consequently, with $\kappa = 997/15000$,

$$\kappa nD + n(C - C_{\mathrm{E}})\rho \le K_F + CA(\rho) + F_\varepsilon(C)|H| - \{q(C, \rho) + e_N(\rho)\}H^2. \tag{A.18}$$

The first unseeded pass uses $e_N = 0$.

The own-score interval is also conservative at N. If a physical cell forces $D \geq d_l \geq 0$ and $\rho \geq \rho_l > 0$, then

$$C \leq .40974 + \frac{r/N - L_N d_l/6}{\rho_l} \tag{A.19}$$

whenever the numerator is nonnegative. A negative numerator excludes the cell. This follows from (6.3), since $r/n - \varphi d_l/6 \leq r/N - L_N d_l/6$. The independent moment envelope and c_{in} may be intersected with (A.19).

For each candidate freeze all physical coefficients before taking the score and moment corners. For fixed H, the right side of (A.16) is convex in c and in ρ separately: A is convex, F_ε is a maximum of affine functions, and q is affine in c and concave in ρ. The same is true after adding the affine curvature e_N. Hence the whole physical rectangle is bounded by its corners, first pointwise in H, and then by completing each corner's square. For $Q > 0$, $\sup_{h \geq 0}(A + Fh - Qh^2) = A + (F_+)^2/(4Q)$; using $A + F^2/(4Q)$ is a weaker valid bound.

For a charged candidate split the score interval at $.40974$ and use

$$n(C - C_{\mathrm{E}})\rho \geq N(C - .40974)_+\rho. \tag{A.20}$$

On either side of this breakpoint the subtracted right side is affine in each of c, ρ separately, so the same corner argument applies. Omitting this breakpoint would not follow from convexity. If R_N is the maximum raw corner price and B_N the maximum charged corner price, the resulting conclusions are

$$\mathcal{R}_n \leq R_N, \qquad nD \leq B_N/\kappa. \tag{A.21}$$

They remain true after minima with other complete candidate bounds. The phase roots of these same corner quadratics bound the original H, and the pointwise charged quadratics give $D \leq \max_j(A_j + F_{j,\varepsilon}|H| - Q_j H^2)/(\kappa n)$. Their normalization is therefore $15000/(997n)$. For an unpaired export the implementation may instead use an already proved $|H| \leq H_*$ and the scalar weighted defect bound to write

$$\kappa n D \leq \kappa N d \leq \kappa N d + \epsilon(H_*^2 - H^2), \qquad \epsilon > 0.$$

This is a valid pointwise envelope on the acquired phase interval. Its artificial quadratic term asserts no additional physical curvature.

A.4. The unseeded step and finite induction.

PROPOSITION A.3 (Fixed-floor interpretation). *Consider the fixed finite B0 or B1 acquisition schedule at $N = 25$. At each Fourier step retain a complete candidate satisfying Lemmas A.1 and A.2 at the floor, its proved physical coefficient bounds* (A.13)*, and the positive corner divisors above. At each physical step retain only necessary inequalities for the current law and a complete closed cover of its incoming domain. Every support or strip implication is proved at the actual integer n, using its own incoming moment and threshold conditions before its output is used. Then every saved raw and defect output (r_j, d_j) has the interpretation*

$$\mathcal{R}_n \leq r_j, \qquad nD \leq N d_j. \tag{A.22}$$

The physical moment, endpoint, sign and threshold bounds exported alongside it apply simultaneously to the same actual-n law.

PROOF. For the first pass there is no input r_j or d_j. Proposition B.2 supplies the initial radii $12/5$ on B0 and $13/5$ on B1 uniformly in the actual integer. Before any defect feedback, $M_4 \le R\rho$ gives the respective ceilings $66/25$ and $78/25$; $M_5 \le RM_4$, $M_6 \le RM_5$ and $|\gamma| \le \rho \le 6/5$ license the triangle low bound and the universal or Cantelli high bound. The initial signed-moment cover uses $\varphi \in [0, 2/5]$, the independent score interval and physical moment inequalities. It assumes neither a central gap nor a final conditional moment cap.

The selected first raw cover maxima, with outward arithmetic, are

$$r_1^{\mathrm{B0}} = 109743487/50000000, \qquad r_1^{\mathrm{B1}} = 283144217/100000000.$$

These are finite upper bounds on the raw prices in (A.16); their closed-domain arithmetic is part of the companion calculation. Both are below $3 < .4N \le nC\rho$. It follows from (6.3) that

$$\frac{\varphi}{6}(\gamma + 3J) = C\rho + \sqrt{n}\mathcal{L} - \mathcal{R}_n/n > 0.$$

Consequently $C_{\mathrm{E}}\rho - \phi(0)D/6 > 0$. Removing the Gaussian weight in (6.3) now lowers its left side and proves

$$n(C - C_{\mathrm{E}})\rho + \kappa nD + n^{3/2}\mathcal{L} \le \mathcal{R}_n. \tag{A.23}$$

Only at this point is the charged output from the same first cover used. Its corner calculation, with $e_N = 0$ and (A.20), gives $d_1 = B_N/(\kappa N)$, with the saved outward values

$$d_1^{\mathrm{B0}} = 129264343/10^8, \qquad d_1^{\mathrm{B1}} = 16619657/10^7.$$

Thus $nD \le Nd_1$, not merely $D \le d_1$. Both $d_1 < 2 < (\sqrt{10} - 1)\rho_l$, so the next pass may use the central gap. This order explains why calculating raw and charged corner arrays in one routine is not a circular use of a defect cap.

Suppose (A.22) and the physical bounds have been established at a later incoming stage. Since $n \ge N$, the weaker physical statement $D \le d_j$ is available. All moment, common-mean, gap and conditional-variance deductions can therefore use d_j in exactly their displayed physical directions. For example, the additional p-local mean caps are

$$\mu^2 \le p(1 + \beta)/2, \quad \mu^2 \le (1 - p)(1 - \beta)/2, \quad \mu^2 \le p(1 - p).$$

They follow from signed-class Cauchy–Schwarz and retain the same meaning on both probability halves. The endpoint-product distances in (A.11) and all signed Pearson rectangles above are likewise independent of n.

The physical inputs now license the next fixed Fourier candidates. Their exact low and high expressions at the actual n are bounded by their floor-N values, by the two lemmas. The floor density interval (A.14) contains the actual density; the floor own-score interval (A.19) contains the actual score; and (A.17) is a lower price for the actual conserved phase. Equations (A.16) and (A.18) therefore hold pointwise in the actual H. Their whole-cell corner maxima yield the next $\mathcal{R}_n \le R_N$ and $nD \le B_N/\kappa$. Writing the latter output as $d_{j+1} = B_N/(\kappa N)$ restores the weighted invariant at every charged stage.

An excluded physical cell has an empty necessary interval or a strictly negative upper bound for a nonnegative resource. Any surviving law lies in at least one retained closed cell, so maximizing its new bounds over all retained cells is valid. Intersections with preceding bounds, finite minima of valid candidates and outward rounding preserve the invariant. This proves the assertion by induction over the prescribed finite schedule, including repeated local INITIAL covers, the resource subiterations and the final exports. The proof invokes neither convergence of the iteration nor extremality at a smaller sample size. □

A.5. Actual support and the interpretation of geometry. The support hypotheses in Proposition A.3 use the actual-index Gaussian majorant (6.1). The original contact function $\mathcal{D}_n$ is zero on the full support; the majorant need only be nonnegative there. Write $s_n = \sqrt{n/(n-1)}$ and $e_n = n(s_n - 1) - 1/2$. The possible extra allowance is $n\rho s_n(C_{n-1} - C)_+ = 0$ under the assumed full predecessor. A support ray or an excluded strip is obtained by proving the right side strictly negative throughout that set, uniformly in all incoming moment, score and threshold parameters. The Gaussian differences, s_n and e_n in that calculation use the actual n. For a bounded H test, that bound on H must already have been proved by an earlier phase envelope. In particular the estimate $e_n < .384/n$, proved in the retained derivation only for $n \geq 36$, is not used in these five actual-index requests.

The initial signed-moment interval follows from the support-free inequality (B.4); it requires no radius or higher-moment feedback.

At the first support call the only moment restrictions are the specified rho band and its consequences, together with the actual score cap. An intermediate call consumes the moment intervals which have already been proved by the preceding resource stage. Its new radius licenses moment improvements only after the actual-index strict contact test has passed. A final moment cap cannot license the first support call, and a successful support calculation at 25 cannot license a support or strip request at $26, \ldots, 30$. This is the one kind of acquisition step for which the floor-index substitution is deliberately not made.

The distinction between weighted and physical quantities is also necessary when exporting noise. From an acquired physical inequality

$$D \geq b + g\delta + h_N N s, \quad \delta = a_0 + a_1, \quad b \geq 0, \quad g, h_N > 0,$$

and $nD \leq Nd_j$, put $r = N/n$ and $R_j = d_j - b$. If $R_j > 0$, $d_0 = R_j/g$ and $L = R_j/h_N$ give (A.24)

$$D \leq r d_j, \qquad D \geq b + g\delta + r h_N \lambda_n, \qquad \frac{\delta}{r d_0} + \frac{\lambda_n}{L} \leq 1 - (n/N - 1)b/R_j \leq 1, \quad \lambda_n = ns.$$

This proves the units of the actual defect, inward intercept and noise price. The joint intercept L stays fixed. Without dividing by R_j, the original inequality reads $g\delta + r h_N \lambda_n \leq r d_j - b$. A negative right side excludes the cell, and a zero right side forces $\delta = \lambda_n = 0$. This also specifies the cases $R_j \leq 0$: zero noise is retained only when the actual scaled right side is zero.

In the positive-residual case $R_j > 0$, suppose physical support also gives $s \leq B\delta$, $B > 0$, and $p(1-p) \geq v_l > 0$. Retain first the full scaled plane in (A.24) and put

$$c = 1 - (n/N - 1)b/R_j, \qquad H_N = \left(\frac{1}{L} + \frac{1}{NBd_0} \right)^{-1}.$$

The same scalar elimination as in (6.14), now using $\delta \geq \lambda_n/(nB)$, $h = r h_N$, $\overline{D} = r d_j$ and $nr = N$, gives

$$\lambda_n \left(\frac{1}{L} + \frac{1}{NBd_0} \right) \leq c, \qquad \lambda_n \leq c H_N. \tag{A.25}$$

This is the stronger scalar export used by the actual-index adapter. For a nonempty cell $c \geq 0$, and $b \geq 0$ gives $c \leq 1$; with this same support constant B, H_N is therefore also a conservative scalar cap. At $c = 0$ the scaled plane retains $\delta = \lambda_n = 0$.

If only the deliberately weakened, unscaled joint plane $\delta/d_0 + \lambda_n/L \leq 1$ is retained, it gives

$$\lambda_n \leq \left(\frac{1}{L} + \frac{1}{nBd_0} \right)^{-1}, \qquad \theta \leq \left(1 + \frac{nv_l}{L} + \frac{v_l}{Bd_0} \right)^{-1} \leq \left(1 + \frac{Nv_l}{L} + \frac{v_l}{Bd_0} \right)^{-1}.$$

These weaker premises license the displayed actual-n scalar expression and the transported variance fraction. They alone do not license replacing that scalar expression by H_N. The stronger cap in (A.25) is available when the full scaled plane is retained. Both harmonic expressions are separate scalar consequences; the original simultaneous intercept in (A.24) remains L. All further signed refinements may use the transported physical θ and the actual support, followed by a new application of the same joint-defect inequality. This completes the mathematical transfer needed for the five actual-index acquisition schedules.

APPENDIX B: GAUSSIAN CONTACT AS A PEANO REMAINDER

The initial support bounds admit a single analytic proof. The key is to recognize the contact kernel as a Gaussian Hermite remainder, and then apply the classical Peano-kernel representation. This replaces the initial actual-index support calculations and supplies the radial contractions used by the tail route. Remaining actual-index refinements and coordinate-strip exclusions retain their own proofs.

Let Z be standard normal. For $g \in C^3(\mathbb{R})$ with bounded third derivative, put

$$L_x g = g(x) - \mathbb{E}g(Z) - x\mathbb{E}g'(Z) - \frac{x^2-1}{2}\mathbb{E}g''(Z).$$

These expectations exist because g, g', g'' grow at most cubically, quadratically and linearly.

LEMMA B.1 (A uniform Gaussian Peano kernel). *For $|x| \geq 2$,*

$$|L_x g| \leq \|g'''\|_\infty \left\{ \frac{|x|^3 - 3|x|}{6} + \frac{3}{4} \right\}. \tag{B.1}$$

PROOF. Taylor's integral remainder anchored at zero, followed by Fubini, gives

$$\begin{aligned} L_x g &= \int_{\mathbb{R}} g'''(t) K_x(t)\, dt, \\ K_x(t) &= \frac{(x-t)^2}{2}\{\mathbf{1}_{\{t<x\}} - \overline{\Phi}(t)\} + (t/2 - x)\phi(t). \end{aligned} \tag{B.2}$$

Indeed L_x annihilates all quadratic polynomials. Applying it to $(u-t)_+^2/2$, and using the first two truncated Gaussian moments, gives exactly this kernel. The anchored remainders are bounded by Gaussian absolute moments of orders at most three, justifying the interchanges. The displayed kernel has integrable Gaussian tails. Testing $g(u) = u^3/6$ and reflecting give

$$\int K_x(t)\, dt = \frac{x^3 - 3x}{6}, \qquad K_{-x}(-t) = -K_x(t) \quad \text{almost everywhere.}$$

Take $x \geq 2$. We first show $K_x(t) \geq 0$ for $t \leq x/2$. For $t = -y \leq 0$, this is the Mills inequality

$$m(y) := \frac{\overline{\Phi}(y)}{\phi(y)} \geq \frac{2x+y}{(x+y)^2}.$$

Cauchy–Schwarz for the integrals of $1, u, u^2$ against $e^{-yu-u^2/2}du$ gives $m^2 + ym \geq 1$. Hence

$$m(y) \geq \frac{\sqrt{y^2+4} - y}{2} \geq \frac{y+4}{(y+2)^2} \geq \frac{y+2x}{(y+x)^2}.$$

The middle comparison follows by squaring positive quantities; the last follows by differentiation in $x \geq 2$.

For $0 \le t \le 1$, $K_x(t)$ increases in $x \ge 2$, since $\partial_x K_x = (x-t)\Phi(t) - \phi(t) > 0$. The elementary bounds $39/100 < \phi(0) < 2/5$, $e^{-u} \ge 1-u$ and $e^{-u} \le 1-u+u^2/2$ give $K_2(t) \ge P(t)$, where

$$P(t) = \frac{1}{5} - \frac{t}{50} - \frac{13t^2}{100} - \frac{7t^3}{200} + \frac{3t^4}{100} - \frac{3t^5}{400}.$$

On $[0,1]$, $P'(t) \le -49t/200 - 1/50 < 0$, so $P(t) \ge P(1) = 3/80 > 0$. For $1 \le t \le x/2$ the same monotonicity in x gives

$$K_x(t) \ge K_{2t}(t) = \frac{t}{2}\{t\Phi(t) - 3\phi(t)\} > 0.$$

The expression in braces increases for $t \ge 1$; at 1 it is positive because $\Phi(1) \ge 33/40$ and $\phi(1) < 49/200$.

Writing $a = x/2$, we can now bound the negative part by

$$\begin{aligned} \int K_x^- &\le \int_a^{2x} (x - t/2)\phi(t)\,dt + \frac{1}{2}\int_x^\infty (t-x)^2\overline{\Phi}(t)\,dt \\ &\le \frac{3}{2}\phi(a) + \frac{1}{2}\phi(2x) + \frac{\phi(x)}{x^4} < \frac{5943}{16000} < \frac{3}{8}. \end{aligned} \tag{B.3}$$

The first integral uses $\overline{\Phi}(a) \le \phi(a)/a$. The second is $\mathbb{E}(Z-x)_+^3/6$, at most $\phi(x)/x^4$ by $\phi(x+u) \le \phi(x)e^{-xu}$. The last rational bound uses $\phi(1) < 49/200$, $\phi(2) < 11/200$ and $\phi(4) < 1/1000$. Each follows from a positive partial sum of the exponential series. Thus $\|K_x\|_1 = \int K_x + 2\int K_x^-$ proves (B.1).

□

The signed moment needed for contact also has a support-free bound. For $a = \mathbb{E}|X| > 0$, $\beta = \mathbb{E}[X|X|]$ and $\rho = \mathbb{E}|X|^3$,

$$a\rho \ge 1 + \beta^2, \qquad a^2 + \beta^2 \le 1, \qquad |\beta| \le \sqrt{\frac{\rho\sqrt{\rho^2+8} - \rho^2 - 2}{2}}. \tag{B.4}$$

For the first inequality, apply Bessel's inequality to $|X|^{3/2}$ and the orthogonal functions $|X|^{1/2}$ and $\operatorname{sgn}(X)|X|^{1/2}$. Their squared norms are a and their inner products with $|X|^{3/2}$ are $1, \beta$. Cauchy–Schwarz for X and $|X| - a$ proves the second inequality; eliminating a proves the last.

PROPOSITION B.2 (Initial support without index subdivisions). *Let (P, H) be a standardized positive closed global maximizing pair at index $n \ge 25$, with $C = C_n \ge C_{\mathrm{E}}$ and $C_{n-1} \le C$. Then*

$$\rho \le 11/10 \implies \operatorname{supp} P \subset (-12/5, 12/5),$$

$$\rho \le 6/5 \implies \operatorname{supp} P \subset (-13/5, 13/5).$$

No higher moment or defect bound is required.

PROOF. For $g(u) = n^{3/2}\Phi((H-u)/\sqrt{n-1})$, Gaussian convolution gives

$$\begin{aligned} \mathbb{E}g(Z) &= n^{3/2}\Phi(H/\sqrt{n}), \\ \mathbb{E}g'(Z) &= -n\phi(H/\sqrt{n}), \qquad \mathbb{E}g''(Z) = -H\phi(H/\sqrt{n}). \end{aligned}$$

Thus $L_x g = \mathcal{H}_n(H, x)$ exactly. Also $\|g'''\|_\infty = (n/(n-1))^{3/2}\phi(0) < 213/500$ for $n \geq 25$. For $s_n = \sqrt{n/(n-1)}$ and $e_n = n(s_n - 1) - 1/2$,

$$0 < e_n < \frac{2}{5n} \leq \frac{2}{125}. \tag{B.5}$$

Indeed, for $z = 1/n \leq 1/25$,

$$(1-z)(1 + z/2 + 2z^2/5)^2 - 1 = z^2\{1/20 - 13z/20 - 6z^2/25 - 4z^3/25\} > 0.$$

The bracket decreases and is positive at $1/25$; omitting the quadratic correction proves the lower bound.

By (B.4), $\rho \leq 11/10$ gives $|\beta| < 127/500$, and $\rho \leq 6/5$ gives $|\beta| < 44/125$. For either $(\rho_u, b, R) = (11/10, 127/500, 12/5)$ or $(6/5, 44/125, 13/5)$, put

$$Q(r) = -r^3 + 3\rho_u r^2/2 + 3br + 2\rho_u/125,$$

$$V(r) = \frac{213}{500}\{(r^3 - 3r)/6 + 3/4\} + \frac{819}{2000}Q(r).$$

At $r = R$, $Q < 0$, $Q' < -8$ and $Q'' < -11$; since $Q''' = -6$, the multiplier of C remains negative on the whole ray. The contact inequality (6.1) therefore has right side at most $V(r)$ for $|x| = r \geq R$, using $C \geq C_{\mathrm{E}} > 819/2000$ in the negative term. Direct substitution gives

$$V(12/5) = -278919/1250000, \qquad V(13/5) = -85993/1250000.$$

In both rows $V'(R) < -5/2$, $V''(R) < -7/2$ and $V''' < -2$. Hence $V < 0$ on both entire rays, contradicting contact at any support point there, including atoms and accumulation points. □

COROLLARY B.3 (All-index radial support at zero predecessor loss). *Let (P, H) be a standardized positive closed global maximizing pair with $n \geq 47$, $C = C_n \geq C_{\mathrm{E}}$ and $C_{n-1} \leq C$. For each row below, the already established moment bounds $\rho \leq \rho_u$, $|\beta| \leq b$ imply* $\operatorname{supp} P \subset (-R, R)$*:*

ρ_u	b	R	*contact upper bound at R*
$7361/4000$	$643/1000$	$96/25$	$-1/20$
$139/100$	$43/100$	3	$-23/100$
$12563/10000$	$1963/5000$	$27/10$	$-9/500$
$603687/500000$	$4014/15625$	$5/2$	$-49/500$

PROOF. The predecessor allowance is zero. The same Gaussian function has $\|g'''\|_\infty < 207/500$ and $e_n < 2/235$ for $n \geq 47$. For each row put

$$Q(r) = -r^3 + 3\rho_u r^2/2 + 3br + 2\rho_u/235,$$

$$V(r) = \frac{207}{500}\{(r^3 - 3r)/6 + 3/4\} + \frac{819}{2000}Q(r).$$

Direct substitution gives the strict upper bounds in the table. In all four rows $Q(R) < 0$, $Q'(R) < -8$, $Q''(R) < -11$, $V'(R) < -5/2$, $V''(R) < -7/2$ and $V''' < -2$. The whole-ray argument in Proposition B.2 therefore applies unchanged. □

These moment bounds are used in predecessor order. The global envelope supplies the first row. The original central preparation, performed at radius 3.84, supplies the last row for all nine surviving central intervals. The complementary acquisition at 3.84 supplies the second row; after contraction to 3, its left-input and signed-resource calculations supply the third. Thus no contraction uses a moment bound obtained from its own output radius. The later coordinate-strip exclusions retain their original bounded-threshold proofs. The unsigned β caps in these resource rows follow from $\beta = (1-\theta)(1-2p) + \theta_+ - \theta_-$, hence $|\beta| \leq (1-\theta)|1-2p| + \theta$; no sign of β is assumed.

APPENDIX C: THE ANALYTIC PROFILE PARAMETERS

The table gives the original joint inward intercept Δ and noise intercept Λ, together with the improved necessary actual-noise cap L_*, for the 34 middle-moment profiles. All entries are exact terminating decimals. The simultaneous signed coefficients, outward support caps and contact domains are supplied as exact rational numbers in the accompanying `profile-parameters.json`; they are acquired conditions on an extremizer, not independent free parameters.

n	interval	Δ	Λ	L_*
26	I_{34}	0.07393	0.448664	0.26204965
26	I_{35}	0.078697	0.469161	0.27350801
26	I_{36}	0.08286	0.484491	0.28185367
26	I_{37}	0.086366	0.494588	0.29863673
26	I_{38}	0.083933	0.467679	0.28297254
26	I_{39}	0.077665	0.420047	0.2556063
27	I_{34}	0.071192	0.448664	0.25573091
27	I_{35}	0.075782	0.469161	0.26738858
27	I_{36}	0.079791	0.484491	0.2759327
27	I_{37}	0.083167	0.494588	0.29258472
27	I_{38}	0.080824	0.467679	0.27794155
27	I_{39}	0.074788	0.420047	0.25182702
27	I_{40}	0.06347	0.34526	0.20897126
28	I_{34}	0.068649	0.448664	0.25004307
28	I_{35}	0.073076	0.469161	0.26192751
28	I_{36}	0.076941	0.484491	0.27068902
28	I_{37}	0.080197	0.494588	0.28726658
28	I_{38}	0.077938	0.467679	0.27351126
28	I_{39}	0.072117	0.420047	0.24847935
28	I_{40}	0.061204	0.34526	0.20687478
29	I_{34}	0.066282	0.448664	0.24487077
29	I_{35}	0.070556	0.469161	0.25700268
29	I_{36}	0.074288	0.484491	0.26599549
29	I_{37}	0.077432	0.494588	0.28254209
29	I_{38}	0.07525	0.467679	0.26957152
29	I_{39}	0.06963	0.420047	0.24548928
29	I_{40}	0.059093	0.34526	0.20498657
30	I_{34}	0.064073	0.448664	0.24012598
30	I_{35}	0.068204	0.469161	0.25252129
30	I_{36}	0.071812	0.484491	0.26175592
30	I_{37}	0.074851	0.494588	0.27830578
30	I_{38}	0.072742	0.467679	0.26603838
30	I_{39}	0.067309	0.420047	0.24279925
30	I_{40}	0.057124	0.34526	0.20327613

The analytic programs use the Python 3 standard library and retain exact rational arithmetic. In the companion directory `finite-steps/source/analytic`, the seven actual-30 profiles can be recomputed with

```
python -B replay_n30_analytic_panels_20261027.py -output
fresh.json
```

An explicit `-output` argument selects a fresh filename; the program refuses to overwrite an existing result. The option `-indices 37` recomputes just that complete profile, including its contact domain and the CDF bridge. The finite-index mathematical companion states the count and phase composition for every profile; the executable calculation supplies the enclosed faces, derivatives and strict margins used there.

APPENDIX D: THE SEVEN LOWER-MOMENT COMPARISON PLANS

This appendix gives the complete plans underlying Proposition 7.1. All use the common comparison, curvature and allocation rules of the main text.

LEMMA D.1 (One lower-moment profile). *Under the retained actual-*26 *acquisition and the global predecessor* $C_{25} \le C_{\mathrm{E}}$*, the entire original profile* $p \in [.40, .41]$*,* $1 \le \rho \le 11/10$ *has a strictly negative conditional count residual. A uniform upper bound is* $-5610547/10^{12}$.

PROOF. The original inward and noise intercepts are $\Delta = .057809$ and $\Lambda = .301161$. The necessary signed resource plane improves the actual-noise cap to $L_* = .17805629$. The original contact inequalities exclude 25 of the 27 count roots, leaving only 10 and 11.

Current-count comparisons cover count 10 on $[-1/2, 0]$ with slope multiplier 1, and count 11 on its two half intervals with multipliers $3/2$ and 1. Their upper bounds are respectively $-.000005610547$, $-.000934723659$, and $-.003337615615$.

For the positive half of count 10, retain the seven noise intervals and four allocation intervals above. On the first noise interval join the current price to the reflected-square price at $t = 1/32$; on the second join at $1/8$; on intervals 3–7 join at $1/4$. Each reflected-square price uses reference count 11, width $h = 7/8$ and Corollary 4.3. Here $11 > 26p$, $p < 1/2$, and the positive affine inward cap decreases in allocation. The moment floor therefore increases in allocation. All next-atom, legal-width and same-noise derivative gates hold on their whole stated domains. In the first interval $y_{\max} = 3/32 > 0$ is allowed by the extended variance-derivative bound in Section 3.

Every current prefix is at most $-.000025632850$; every clipped suffix is at most $-.001597137326$. Probability and phase chords include every closed boundary. The two pieces in each noise/allocation region meet at the stated phase, and the full noise union is $[0, L_*]$. Together with the three half-interval comparisons and the contact exclusions this covers the whole original profile. The least strict bound is $-.000005610547$. □

This replaces a conditional count comparison. The original support, floor and acquisition proofs remain its inputs; replaying the pure physical acquisition formulas does not establish those inputs anew.

LEMMA D.2 (An adjacent lower-moment profile). *Under the retained actual-*26 *acquisition and* $C_{25} \le C_{\mathrm{E}}$*, the entire original profile* $p \in [.39, .40]$*,* $1 \le \rho \le 11/10$ *has conditional count residual at most* $-98710877/10^{12}$.

PROOF. The original intercepts are $\Delta = .05769$, $\Lambda = .30708$; the same signed resource argument gives $\lambda \le .17375152$. Contact excludes 24 count roots. Only $9, 10, 11$ remain, and the count-9 contact lower phase is greater than $1/4$. Count 11 has the two current-count half-interval comparisons with slope multipliers $3/2, 1$.

For the lower counts one reference law covers both sides of a nearest-count boundary. Use source count 9, reference count 10 and width $7/8$. On noise intervals $1, 2$ its auxiliary phase runs through $[1/4, 7/8]$, and on intervals 3–7 through $[1/4, 3/4]$. The exact threshold identity

$$9 + (1 + t) = 10 + t$$

therefore covers count 9 and the lower part of count 10 by one continuous price in each noise group. Lemma 2.1 holds for every real phase, so an auxiliary phase greater than $1/2$ is permitted. Here $10 < 26p$ throughout the probability interval, and Corollary 4.3 selects the lower allocation endpoint for both variance and moment cost.

The remaining count-10 negative phases are $[-1/8, 0]$ on the first two noise intervals and $[-1/4, 0]$ on the others; current-count prices cover them. Its positive half uses the same $1/32, 1/8, 1/4$ handoffs as in Lemma D.1, now with this profile's own resources. Reference count $11 > 26p$ selects the upper allocation endpoint for variance and the lower one for moment cost. All legal-width, next-atom, noise-derivative and continuous probability/phase gates are strict on their stated domains. The whole closed unions retain every original count, noise and allocation region, with largest upper bound $-.000098710877$. □

LEMMA D.3 (Reversed signed noise prices). *Under the retained actual-*26 *acquisition and* $C_{25} \le C_{\mathrm{E}}$*, the entire original profile* $p \in [.41, .42]$*,* $1 \le \rho \le 11/10$ *has conditional count residual at most* $-24705293/(5 \cdot 10^{11})$.

PROOF. The original intercepts are $\Delta = .054236$, $\Lambda = .276951$. Here $h_+ - h_- < 0$. The positive endpoint coefficients in (6.13) still give a valid finite resource iteration and $\lambda \le L_* = .162619$, retaining the original joint intercept Λ. For the actual moment floor, (4.10) is greater than $.473372164977$; both negative terms from the increasing inward cap are included. Thus G increases with allocation on the whole domain.

For current-count prices, the necessary weaker plane

$$g(a_0 + a_1) + \lambda h_- \le \mathfrak{R} + L_*(h_- - h_+)_+$$

supplies an allocation-independent inward cap. Its intercept is smaller than the original Δ, and its noise slope is larger than Δ/Λ, as required by the fixed-reference estimate.

Contact excludes 24 count roots and leaves $10, 11, 12$. Count 10 starts above $-1/8$, and count 12 ends below $-1/4$. Current-count prefixes cover count 10 through $1/32, 1/8, 1/4$ on the first, second and remaining noise intervals. One source-10, reference-11 square then covers both its positive suffix and the negative prefix of count 11. On noise intervals $1, 2$ the width is $7/8$ and its auxiliary phase ends at $7/8$. On intervals 3–6 use width $9/8$ with phase $[1/4, 7/8]$. On the last interval use width $7/8$ and phase $[1/4, 3/4]$. The identity $10 + (1 + t) = 11 + t$ gives the original count-11 phases exactly.

Current-count prices complete count 11 through zero and on its positive half; count 12 uses slope multiplier $3/2$ on $[-1/2, -1/4]$. The original seven noise intervals and four allocation intervals are unchanged. All width, next-atom, allocation and noise gates, followed by the probability/phase chords, give a strict bound on the full closed unions. Their largest upper bound is $-.000049410586$. □

COROLLARY D.4 (A complete lower-moment strip). *At each integer* $n = 26, \dots, 30$*, under the retained actual-*n *acquisition and* $C_{n-1} \le C_{\mathrm{E}}$*, the original profile* $p \in [.41, .42]$*,* $1 \le \rho \le 11/10$ *has the following conditional count bounds. The original noise intercept is* $\Lambda = .276951$ *in all rows. The five residual bounds are the corresponding rows of Proposition 7.1.*

PROOF. Use the same comparison rules with each integer's own masses, variance and moment normalization. Lemma D.3 gives 26. At 27, contact excludes 25 of the 28 roots and leaves $10, 11, 12$. Reference count $11 < 27p$ uses lower allocation endpoints for variance and cost, while $12 > 27p$ uses the two allocation corners. The signed moment derivative gate is positive. On the lowest noise interval the source-10 square runs through auxiliary phase

$15/16$, which is exactly count-11 phase $-1/16$. The current price on $[-1/16, 1/32]$ and its clipped-square continuation then close the middle count.

At 28, the remaining counts are 11 and 12. Their outer phase intervals are $[-17/64, 1/2]$ and $[-1/2, 15/32]$. The same fixed current references for count 11 use the phase faces $-1/8$ and 0, followed by the square continuations. Count 12 joins current slopes $3/2$ and 1 at phase $-1/32$. Reference count $12 > 28p$ has fixed variance direction throughout the probability interval.

At 29, contact excludes 27 roots and leaves $11, 12, 13$. Reference count 12 crosses $29p$ at $p = 12/29$ inside the original probability interval. Use both variance-endpoint branches in (4.8), with the same actual noise and the lower-allocation moment cost in both. Each branch has its own whole-interval noise and curvature checks before their maximum is bounded. The lowest-count square starts at phase $1/8$, below the contact lower bound .15468890249; on the sixth noise interval, auxiliary phase $13/16$ meets original count-12 phase $-3/16$.

At 30, contact leaves only $12, 13$, with valid outer phase intervals $[-7/16, 1/2]$ and $[-1/2, 5/16]$. The negative part of count 12 uses a lower-allocation square and current prices; on the lowest noise interval the same current reference is interpolated through one extra phase face, $-1/8$. Its positive part hands off at $1/128$ with width $2/3$ on that interval. Here $y_{\max} = 125/384 > 0$ and the one-clip and variance-derivative conditions hold on the whole region. Higher-noise square prices use width $5/4$ where specified by the plan, and the shifted-count joins close count 13.

All these explicit finite plans retain the seven original noise and four allocation intervals. Their remaining pieces use the current or square envelopes, and their closed unions cover every retained contact interval. Their largest upper bounds are those in the table. □

APPENDIX E: LOCATIONS OF THE COMPANION PROOFS

The moment-defect and original count arguments are due to Rodziewicz–Kołodziejek [8]; the completion notes identify their reconstruction scope. The companion sources retain their original attribution, theorem labels and statement scopes. A label quoted below belongs to the named companion, not to the numbering of this article. The source collection is prepared by `prepare_global_companions.py`; it includes the local TeX inputs of the multifile large-index proof.

E.0.0.1. Input (I).. In `companions/low-index/`, the first-index bound is `n1/n1_base.tex`, Theorem 1 (`thm:base`). The source `n2-n3/manuscript.tex` contains `n2:thm:main` and `n3:thm:main`, together with their support reductions and continuous-domain polynomial certificate implications. The complete fourth-index proof is available in the existing public manuscript PDF `companions/references/rk-v3/moment-defects-v3.pdf`, Section 26, Theorem 26.1 and its proof. A TeX source for that complete fourth-index proof is not included.

The subsequent compositions are Theorem 1 (`composition`) in `n5/five_sum_completion_20261025.tex`; Proposition 1 and its completed composition in each of `n6/six_sum_interface_20261025.tex`, `n7/seven_sum_interface_20261025.tex`, `n8/eight_sum_interface_20261025.tex`, and `n9/nine_sum_interface_20261025.tex`; the theorem entitled *Unrestricted tenth-index conclusion* in `n10/main.tex`; and the ordered implication at the end of `n11-14/conditional_11_14_interface_20261026.tex` (which has no separately numbered conclusion). The final blocks are Theorem 1 in `n15-19/main.tex`, Proposition 1 and its completed composition in `n20/twenty_sum_interface_20261025.tex`, Theorem 1 in `n21-24/main.tex`, and Theorem 1 in `n25/main.tex`. The high-moment complement at the actual indices 25–30 is stated and derived in `n25-fourier/main.tex`.

These interfaces are accompanied by continuous derivations. The source `companions/common/finite-analytic/BE_cutoff35_streamlined.tex` contains `global:attainment`, `global:contact`, `complete44ent:smooth-fourier`, `complete44ent:fourier-early-low`, `complete44ent:re-enclosure`, `res:contract`, and the section *One finite-count comparison for all the new indices*. The additional small-index proofs retain their original text format under `companions/low-index/continuous-sources/work/`. In particular, the fifth-index `cutoff5_20261020/audit/ANALYTIC_DEPENDENCIES_V2.txt` locates the conserved-identity, finite-window LOW, resource and count derivations; their mathematical proof texts are included alongside that source map. They are distinct from numerical success records. The original frozen arithmetic inputs and actual returned objects are retained under `companions/low-index-support/`. Its `read_low_index_support.py` describes the existing joins, their exact input locations and their Python or compiler requirements.

E.0.0.2. Input (II).. In `companions/actual-index/`, the coarse entry from $C_{30} < C_{\mathrm{E}} + .01$ and the literal bridges to 34 are Propositions `prop:31` and `prop:bridges` in `n31-34/main.tex`. The exceptional band-7 defect-domain argument is `n31-domain/sharp31_B7_D_domain_20261025.tex`. Continue with the theorem entitled *Unrestricted index–35 conclusion* in `n35/actual35_conditional_composition_20261025.tex`, the proposition entitled *Unrestricted conclusions at 36, 37 and 38* in `n36-38/actual36_38_conditional_composition_20261026.tex`, and Propositions 1 and 2 in `n39/main.tex`. The actual-39 continuous-domain extension is separately included in `n39-continuous/actual39_continuous_lemmas_20261026.tex`. Theorem `thm:main` in `n40-48/main.tex` contains the bridge through 48; the present composition uses its steps only through 46. Its two local implications are `prop:forty` and `prop:two`. The directories `n40-continuous/`, `n40-original/`, `n42-continuous/`, and `n42-local/` contain the corresponding actual-index and finite-band continuous arguments. The corresponding compressed inputs, returned objects and existing join programs are retained under `companions/bridge-support/`. The accompanying `read_bridge_support.py` locates the mathematical objects and launches the original joins against those collected inputs.

E.0.0.3. Input (III).. The local resource implication `re:localG` in `paper_entry/resource_entry.tex` is used with $b \geq 0$, $P \geq 0$, $\ell_- \geq 0$ and $L_0 \geq 0$. These are the physical coefficients of `conf:moment` and the nonnegative noise subintervals; they license the substitutions $nP \geq NP$ and $n\theta \geq \ell_-^2 L_0$.

The proposition `global:local` in `companions/large-index/analytic-route/paper/sections/global_route.tex` contains the zero-allowance uniform local exclusion (III) used here. In the present proof, Corollary B.3 replaces its initial radius-3.84 calculation over six index bands, both left radial contractions and the central radius-2.5 contraction. The resource schedule, coordinate-strip exclusions and local comparisons are retained. The broader original heat-expansion route remains in the archive. The continuous analytic route is the multifile source `companions/large-index/analytic-route/paper/main.tex`: `re:acquired`, `re:enclosure`, and `shortloc:theorem` supply its resource acquisition and local comparison. Its included coefficient atlas derives compensated inversion and the conserved identity, rather than treating them as numerical premises. The independently completed $n \geq 49$ route is also retained in `companions/large-index/completion/n49_completion_20261025.tex`; the new induction from $45, 46$ uses the local exclusion directly. The selected compressed source/input archive and its original returned calculation archive are included under `companions/uniform-tail/`. The accompanying `prepare_uniform_tail_support.py` can unpack them

to a new directory, reconstruct the exact local cover, and run the original transcript-to-target join. Its Windows schedule launcher changes two path-key expressions to POSIX spelling; all mathematical conditions and original archive contents are preserved.

E.0.0.4. Weighted localization and the 34 finite profiles.. Appendix A gives the mathematical proof of (6.15), including the raw first step, all LOW and HIGH terms, the weighted induction and the order of actual-index support acquisition. The earlier fixed-floor derivations are retained in `companions/localization/analytic/UNIFORM25_CANTELLI_PRODUCTGAP_REVIEW.txt` and `companions/localization/analytic/UNIFORM25_BETA0_SOURCE_AND_COMPOSITION_REVIEW.txt`. The inward-plane scaling is derived in `companions/localization/original/new_audit/SCALED_INWARD_PHYSICAL_DERIVATION_V1.txt`. Their continuous foundation is the resource-enclosure and simultaneous-contract argument identified above; actual-index support acquisitions are still separate inputs.

The initial B0/B1 radii now follow directly from Proposition B.2. Their original actual-index radius computations are no longer premises of this first step; later support refinements still use their own incoming data. The Peano-kernel and rational-ray calculations are also retained in `gaussian_contact_peano_20261028.tex` and `gaussian_contact_peano_20261028.py` in the finite analytic directory.

The full finite comparison text is `companions/finite-steps/source/analytic/finite_gap_shared_stoploss_20261026.tex`. Theorems `global-c26-complete`, `global-c27-complete`, `global-c28-complete`, `global-c29-complete`, and `global-c30-complete` give the five actual-index compositions. Their proofs identify the constituent profile lemmas. At 26, these include `p34-p35-count8-complete`, `p35-count9-second-moment-complete`, `p36-p37-count10-second-moment-complete`, and the complete-panel corollaries for 36–39. At 27, the profile conclusions are `n27-p34-complete`, `n27-p35-complete`, `n27-p36-complete`, `n27-central-complete`, and `n27-p39-complete`, `n27-p40-complete`; the central corollary covers both 37 and 38. The corresponding 28 statements are `n28-p34-p35-complete`, `n28-p36-complete`, `n28-p37-complete`, `n28-p38-complete`, and `n28-p39-p40-complete`. The last two groups are the lemmas `n29-seven-panels-complete` and `n30-seven-panels-complete`. The lower-moment replacement in Lemma D.1 is the separate program `probe_b0_shared_reference_20261027.py` in that analytic source directory, with the exact output `result_b0_n26_p40_complete_frozen_20261027.json`. Its `-complete` entry recomputes this whole conditional profile without reading a stored numerical result. Lemma D.2 uses the separate `probe_b0_p39_shared_20261027.py` and `b0_allocation_interval_clip_20261027.py`, with output `result_b0_n26_p39_complete_frozen_20261027.json` in the same analytic directory. The signed-price extension in Lemma D.3 is `probe_b0_p41_shared_20261027.py`, with `b0_two_endpoint_budget_20261027.py` and `b0_signed_allocation_interval_20261027.py`; its result is `result_b0_n26_p41_complete_frozen_20261027.json`. Corollary D.4 reuses these helpers through the explicit `PLAN` in `probe_b0_n27_p41_shared_20261027.py`; the actual-27 output is `result_b0_n27_p41_complete_fresh_20261027.json`. The actual-28 plan is `probe_b0_n28_p41_shared_20261028.py`, with output `result_b0_n28_p41_complete_fresh_20261028.json`. The crossing application at 29 is the `PLAN` in `probe_b0_n29_p41_shared_20261028.py`, using `b0_variance_endpoint_branches_20261028.py`; its output is `result_b0_n29_p41_complete_fresh_20261028.json`. The actual-30 plan is `probe_b0_n30_p41_shared_20261028.py`, with final output

`result_b0_n30_p41_complete_frozen_20261028.json`. Its earlier failed complete candidate is retained in the research sources and supplies no completed profile bound. The finite lower-envelope implementation in `probe_classical_order_compression_20261028.py` replaces optimization over mixture weights. Its exact algebraic replay of all 25 nonempty N27/P35 mixed regions agrees with the old optimized bound. All other old fixed-mixture bounds transfer by the pointwise domination following Lemma 3.2; no numerical regeneration is required for this logical transfer. The acquisition, lower-moment, ordinary-forest and unrestricted Fourier inputs and returned objects are retained separately in `companions/finite-other-support/`. Its `read_finite_other_support.py` provides the original mathematical readers with temporary source materialization. The expensive complete lower-moment reader is an explicit option. Partial historical count runs retain their original domains; the analytic profile comparisons supply the stated complementary regions.

The companion text identifies the continuous inequalities and compositions. The frozen coefficient data, closed subdivision trees, exact arithmetic programs and archived fresh calculations are retained in the collections above. Their mathematical readers relate returned objects to the complete parameter covers and consumed contracts. Fresh numerical regeneration is a separate operation using the retained producer entrypoints.

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